

Uncredited precedence in
 “A spectral gap throughout the high-temperature
 Sherrington–Kirkpatrick phase”
 (OpenAI, September 24, 2026)

October 8, 2026

Summary of findings

Grades and categories are defined in Section 1. Locations refer to the preprint’s compiled numbering. “Credited?” names the earlier work and says what the preprint cites for the content in question.

ID	Grade	Technique or result	In the preprint	Credited?
S1	STRONG	Kac–Rice complexity of TAP critical points in planted SK: gradient density, determinant rate, relative-entropy rate with its zero at the Bayes point; the count restricted to Hessian failures, with a Sudakov–Fernique bound on the smallest Hessian eigenvalue	§5.1–5.4, (5.4)–(5.9), Lemmas 5.3–5.5 (pp. 28–38); Lemmas 3.2–3.5; Lemmas 4.1–4.2	Fan–Mei–Montanari 2018 [1]; Celentano–Fan–Mei 2021 [2]: not cited
S2	STRONG	Approximate-critical-point control from an annealed Kac–Rice bound: smoothing, a local lower bound in which the volume cancels the determinant, and Markov	Lemma 5.6 and the exclusion step in the proof of Proposition 5.1 (pp. 35–38)	Huang–Sellke 2023 [3]: not cited; presented as the preprint’s own
P1	PARTIAL	Stein identity for tanh under a cosh-tilted Gaussian, used to compare a field with an Onsager-corrected field	(6.8)–(6.9) (p. 45); Lemma 6.5 (pp. 53–54); proof of Proposition 7.2 (p. 57)	Chatterjee 2007 [4]: not cited
P2	PARTIAL	Planted SK law under Gaussian observation and its likelihood transfer; state-evolution fixed point	(5.4) (p. 28); §5.4 (pp. 39–40); Lemma 4.2 (p. 25)	El Alaoui–Montanari–Sellke 2022 [5]: cited in §1.1 as a parallel use, not at the points of use
P3	PARTIAL	Route to a Poincaré inequality through covariance bounds at stochastic-localization fields via the inverse TAP Hessian	Proposition 5.1, Proposition 7.2, §2	El Alaoui–Gaitonde 2022 [6]: cited only for a different result (its Theorem 1.1)

ID	Grade	Technique or result	In the preprint	Credited?
P4	PARTIAL	Stopped stochastic localization with exceptional paths	§2.2 and §2.4 (pp. 7–10); Corollary 5.2	Huang–Mohanty–Rajaraman–Wu 2024 [7]: not cited; Chen–Eldan cited for the version without exceptional paths
P5	PARTIAL	Resolvent diagonals controlled uniformly over all diagonal deformations by metric projection, regular increments, concentration and chaining	Lemma 3.1, §3.3 (pp. 17–19)	Davies–Lee–Sandhu–Shi 2026 [8]: not cited
P6	PARTIAL (B)	Weak Poincaré inequality for SK Glauber dynamics at $\beta < 1/2$ via approximate stochastic localization	Theorem 1.1 (p. 3); §1.1 (p. 4)	Davies–Lee–Sandhu–Shi 2026 [9]: not cited; the later cited Wang and Boban–Li–Oveis Gharan both discuss and imply it

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1 About this document

The preprint. The preprint is *A spectral gap throughout the high-temperature Sherrington–Kirkpatrick phase*, dated September 24, 2026, with author line “OpenAI” [10]. It is one of 719 manuscripts in the `openai/math` release, all produced by an unreleased internal OpenAI model. It proves that for every fixed inverse temperature $0 < \beta < 1$, the zero-field Gaussian SK Gibbs measure satisfies, with probability tending to one over the disorder, a Poincaré inequality for the unscaled heat-bath Dirichlet form with a constant depending only on β . Section 2 summarizes the paper and fixes the notation used throughout this document. The preprint cites no other preprint from the same release.

What counts as a finding. This document reports only:

- (A) **Unattributed reuse.** A technique, construction, identity, estimate or body of theory developed in an identifiable earlier paper is reused without attribution: either nothing is cited, or the citation given covers something else or something narrower than what is reused.
- (B) **Same-problem precedence.** Earlier, uncited partial progress on the same problem, such as a weaker or special-case version of a statement the preprint proves.
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Grades.

Strong. The earlier technique is reused essentially intact without credit for that content. This holds even when the preprint combines it with new ingredients or applies it in a new setting. New ingredients are listed as differences and do not lower the grade.

Partial. Only a fragment of the earlier technique is reused, or the credit is present but only indirectly.

Same-problem findings (B) are graded by how close the earlier result comes to the preprint’s: STRONG when it proves essentially the same statement, PARTIAL when it is a weaker, special-case or partial-range version. Structural findings (C) are graded by impact instead:

Major. A route, a section, or a component of the main theorem becomes a citation.

Moderate. A multi-page chain of lemmas becomes a citation.

Grades measure how completely the *earlier technique* is reused, not how much of the preprint’s argument it accounts for. Credit counts as indirect only when the preprint cites a source *for that specific content*, and the cited location itself covers or credits the original. Citing the same paper nearby for a different result does not count. This matters because the preprint states that “we prove the model-specific estimates in the sections below” (§1.2, p. 5), and most of its later sections contain no citations.

How to check a claim. Locations in the preprint use its compiled numbering (sections, equations, theorems, lemmas, propositions) and, where helpful, page numbers in its PDF. Locations in other sources give the theorem, equation and page of the version listed in the bibliography. Each finding ends with a short checklist of the passages to put side by side.

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2 The preprint in brief, and notation

Model and theorem. Fix $0 < \beta < 1$ and write $j = \beta^2$. The disorder J is a symmetric $n \times n$ matrix with zero diagonal and independent entries $J_{ik} \sim N(0, j/n)$ for $i < k$. For a field $h \in \mathbb{R}^n$, the Gibbs law on $\{-1, 1\}^n$ is

$$\mu_h(x) \propto \exp\left\{\frac{1}{2}x^\top Jx + h^\top x\right\},$$

so μ_0 is the zero-field SK model. With P_i^h the conditional expectation under μ_h given all spins but the i th, the unscaled heat-bath Dirichlet form is $\mathcal{D}_h(f) = \sum_i \mathbb{E}_{\mu_h}(f - P_i^h f)^2$ (1.1). The main result (Theorem 1.1) is that, with probability tending to one over J , $\text{Var}_{\mu_0}(f) \leq C_\beta \mathcal{D}_0(f)$ for every $f : \{-1, 1\}^n \rightarrow \mathbb{R}$. The conditional mean and variance of x_i under μ_h are $m_i^1 = \tanh h_i^1$ and $v_i^1 = 1 - (m_i^1)^2$, where $h^1 = h + Jx$ is the local field (§1.2). For a vector v , $\langle v \rangle = n^{-1} \sum_i v_i$.

Route of the proof. A spin sample is observed in Gaussian noise,

$$X \sim \mu_0, \quad Y(t) = tX + B(t), \quad 0 \leq t \leq T,$$

with B a standard Brownian motion (1.4); given the observations up to time t , the spin law is $\mu_{Y(t)}$. This is stochastic localization in its Gaussian-channel form. Section 2 of the preprint transfers three inputs back to μ_0 :

1. stability of the TAP field equation at every field visited by the observation path, except on an event of probability e^{-an} (Proposition 5.1, built on §§3–5);

2. at such stable fields, a directional covariance bound $\|\text{Cov}_{\mu_h}(G, x)\|^2 \leq C(\text{Var}_{\mu_h}(G) + \mathcal{D}_h(G))$ (Proposition 7.2) and a bound on the mean of x under $f^2\mu_h$ (Proposition 7.3), both obtained from finite recursions and residual identities (§6);
3. a terminal gap $\text{Var}_{\mu_{Y(T)}} f \leq 2\mathcal{D}_{Y(T)}(f)$ at a large fixed time T (Proposition 8.3).

The field map and stability. Fields are written in the coordinates $y = \text{arctanh } m$. Put $m(y) = \tanh y$, $q(y) = \langle m(y)^2 \rangle$, $b(y) = 1 - q(y)$ and $V_y = \text{diag}(1 - m(y)^2)$. The field map (5.1) is

$$F_{J,h_0}(y) = y - h_0 + (jb(y)I - J)m(y),$$

whose zeros are the TAP equations with Onsager term $jb m$. For a nonnegative diagonal matrix A , the stability matrix (5.2) is

$$H_J(y, A) = I + A^{1/2} \left(jb(y)I - J - \frac{2j}{n} m(y)m(y)^\top \right) A^{1/2};$$

at $A = V_y$ it is $V_y^{1/2} \mathcal{H}(m) V_y^{1/2}$, where $\mathcal{H}(m) = \text{diag}(1/(1 - m^2)) + j(1 - q)I - J - \frac{2j}{n} mm^\top$ is the Hessian of the negative TAP free energy. “Stable” means that every approximate zero, $\|F_{J,h_0}(y)\| \leq 2\rho_0\sqrt{n}$, has $H_J(y, A) \succ cI$ for all diagonal A near V_y (5.3).

The stability integral. Stability is proved under the planted law (5.4)

$$\tilde{J} = W + \frac{j}{n} \mathbf{1}\mathbf{1}^\top, \quad h_0 = t\mathbf{1} + \sqrt{t}g,$$

with W a GOE matrix of scale j and g standard Gaussian, using the smoothed integral (5.5)

$$I_{t,\sigma}(\mathcal{E}) = (2\pi\sigma^2)^{-n/2} \int \mathbf{1}_{\{(y,\tilde{J}) \in \mathcal{E}\}} \exp\left\{-\frac{\|F_{\tilde{J},h_0}(y)\|^2}{2\sigma^2} + \frac{njb(y)^2}{2}\right\} dy.$$

Its exponential rate is a function of the empirical law P of the coordinates of y : with $M = \mathbb{E}_P \tanh$, $q = \mathbb{E}_P \tanh^2$, $b = 1 - q$, $d = t + jM$, $s = t + \sigma^2 + jq$, $S = s + jq$ and $a = \mathbb{E}_P[(y - d) \tanh y]$, the rate is $\psi - \text{KL}(P\|N(d, s))$ with $\psi = j(a - sb)^2/(2sS)$ (5.6)–(5.9). The preprint’s scalar entropy inequality (Lemma 4.1) states $\text{KL}(P\|N(d, s)) \geq \psi$, with equality only at $P = N(d, s)$; the relevant equality law is $N(s_*(t), s_*(t))$, where $s_* = t + j \mathbb{E}_{N(s_*, s_*)} \tanh$ (Lemma 4.2).

The recursion. At a field h_0 , starting from a spin sample $x \sim \mu_{h_0}$, the preprint runs (6.1)

$$m^0 = x, \quad b_0 = 0, \quad m^l = \tanh h^l, \quad b_l = \langle 1 - (m^l)^2 \rangle, \quad h^{l+1} = h_0 + Jm^l - jb_l m^{l-1},$$

with residuals $R_l = m^{l-1} - m^l$, through a fixed finite depth. Spin-flip derivatives are $d_i f(x) = (f(x) - f(x^{(i)}))/(2x_i)$, where $x^{(i)}$ flips the i th spin, and $\|G\|_*^2 = \mathbb{E}_{\mu_{h_0}} G^2 + \mathcal{D}_{h_0}(G)$ (§1.2).

3 Strong findings

3.1 S1. The Kac–Rice complexity method for TAP critical points in planted SK, with the count restricted to Hessian failures

Summary. The stability proposition of the preprint is proved with the Kac–Rice computation of Fan, Mei and Montanari (“FMM”) [1] for critical points of the TAP free energy in planted SK: the gauge $x = \mathbf{1}$, the TAP gradient, its density from $\text{Cov}(Wm)$, the determinant rate [1, Prop. 3.2,

(3.5), p. 9], and the relative-entropy rate [1, Prop. 3.5 and §3.4, pp. 10–11], which vanishes at the Bayes point [1, Prop. 5.2, p. 22]. As Celentano, Fan and Mei (“CFM”) do [2, Lemmas 4.7–4.9, Cor. 4.10, pp. 17–18], it restricts the count to Hessian failures near the Bayes law, and it uses CFM’s Sudakov–Fernique comparison [2, Lemma 4.1, p. 15] (which CFM apply to the smallest eigenvalue of the TAP Hessian) as its bulk input. In the preprint this is §5.1–§5.2 and the bound (5.25) (pp. 28–37), with Lemma 3.5 (p. 20) for the determinant and Lemma 3.2 (p. 12) for the comparison; at $t = \sigma = 0$ its scalar rate is, term by term, FMM’s variational functional of ρ from §3.4, whose constrained supremum is their complexity. Neither FMM nor CFM is cited, and the words Kac, Rice and complexity do not occur. The only citation on TAP-Hessian positivity is Celentano [11], for local convexity of the El Alaoui–Montanari–Sellke TAP free energy near late AMP iterates (§1.1, p. 4); that paper proves this without the Kac–Rice formula and credits CFM for the analogous result for the FMM free energy and for key steps [11, pp. 4–5]. Lemma 3.2 presents the comparison as a standard Gaussian fact with only Vitale cited. Grade: STRONG, because FMM’s computation is reproduced at the level of equations and CFM’s restricted count at the level of statements, without credit.

Setup. Section 2 defines F , $H_J(y, A)$, V_y , the planted law (5.4), the integral $I_{t,\sigma}$ (5.5) and the scalar parameters $M, q, b, d, s, S, a, \psi$ (we write a for the preprint’s $a_y = \langle (y - d\mathbf{1})m \rangle$). The finding also uses:

- *Gaussian reduction* (Lemma 5.3, (5.8), pp. 28–29). With $z = y - d\mathbf{1} + jbm$ and $\varphi_{d,s}$ the $N(d, s)$ density,

$$\mathbb{E}I_{t,\sigma}(\mathcal{E}) = \int \sqrt{s/S} \prod_i \varphi_{d,s}(y_i) e^{n\psi} \mathbb{P}((y, \tilde{J}) \in \mathcal{E} \mid Wm + \sqrt{t + \sigma^2} g' = z) dy.$$

By Lemma 4.1 the unrestricted rate $\psi - \text{KL}(P \| N(d, s))$ is ≤ 0 (5.9), p. 29; by Lemma 4.2, equality at $\sigma = 0$ holds only on the curve $P_t = N(s_*(t), s_*(t))$.

- *Witness set* (§5.4, p. 37): for $t \geq t_0$, $E_t(W)$ is the set of y for which some diagonal $0 \leq A \leq (1 + \eta)I$ with $\|A - V_y\|_F \leq \epsilon_* \sqrt{n}$ has $\lambda_{\min}(H_{\tilde{J}}(y, A)) \leq c_E$.
- *Determinant* (Lemma 3.5, (3.26), p. 20): for each fixed $0 < \gamma \leq \gamma_0(j, \varepsilon)$, with probability $1 - Ce^{-cn}$, simultaneously over diagonal $0 \leq V \leq I$ and E of bounded rank and norm, $L = I + (jbI - W)V + E$ with $b = n^{-1} \text{Tr } V$ has $n^{-1} \log \det(L^\top L + \gamma I) \leq jb^2 + \varepsilon$; it is applied to $L = F'_{\tilde{J}, h_0}(y)$ (p. 38).
- *Bulk comparison* (Lemma 3.2, (3.8), p. 12): for deterministic diagonal $0 \leq A \leq a_+ I$, $ja_+^2 < 1$, $\chi_A = n^{-1} \text{Tr } A$, $S_A(\tau; W) = I + \tau^2 j \chi_A A - \tau A^{1/2} W A^{1/2} \succeq c_0 I$ for all $\tau \in [0, 1]$ with probability $1 - Ce^{-cn}$. At $\tau = 1$, $A = V_y$ it is $V_y^{1/2}(\cdot) V_y^{1/2}$ of the TAP Hessian minus its rank-one terms.

Role in the argument. Proposition 5.1 (p. 27), input 1 in Section 2, says that along the observation path every approximate zero of F has $H_J(y, A) \succ cI$ for all nearby A . Its proof rests on $\mathbb{E}I_{t,\sigma}(E_t) \leq e^{-a_0 n}$ (5.25), p. 37, which is the Kac–Rice count of this finding: the scalar rate excludes everything outside a neighborhood of P_t (Lemma 5.4, pp. 29–31), and near P_t the conditional Hessian fails with probability at most $e^{-a_* n}$ (Lemma 5.5, pp. 31–35). Passing from (5.25) to approximate zeros (Gaussian smoothing and the local-volume Lemma 5.6) is finding S2, where Lemma 3.5’s rate cancels the weight $e^{njb^2/2}$; the planted law and its transfer are finding P2.

The earlier technique. FMM and CFM study $Y = (\lambda/n)xx^\top + W_F$ with $x \in \{\pm 1\}^n$ and W_F a GOE matrix with off-diagonal variance $1/n$, and the TAP free energy at inverse temperature β ;

CFM take the Bayes (Nishimori) case $\beta = \lambda > 1$. Write $Q = \|m\|^2/n$ and $M = \langle \mathbf{1}, m \rangle/n$. We denote FMM's scalar auxiliary variable (their y) by ϑ , their $A(\rho) = \int x \operatorname{arctanh} x \rho(dx)$ by a_F , and their $L(m)$ by $L_F(m)$.

1. *Gauge and derivatives* [1, §3, (3.2)–(3.3), p. 8]: by sign symmetry $x = \mathbf{1}$, and

$$\begin{aligned} g_n(m) &= \operatorname{arctanh}(m) - \frac{\beta\lambda}{n} \langle \mathbf{1}, m \rangle \mathbf{1} - \beta W_F m + \beta^2(1 - Q)m, \\ H_n(m) &= \operatorname{diag}(1/(1 - m^2)) - \frac{\beta\lambda}{n} \mathbf{1}\mathbf{1}^\top - \beta W_F + \beta^2(1 - Q)I - \frac{2\beta^2}{n} m m^\top. \end{aligned}$$

2. *Kac–Rice integrand*. $\mathbb{E} \operatorname{Crit}_n \leq \int \mathbb{E}[|\det H_n(m)| \mid g_n(m) = 0] p_m(0) dm$ [1, Prop. 3.1, (3.4), p. 9], with $\mathbb{E}[|\det H_n(m)| \mid g_n(m) = 0] \leq e^{nL_F(m) + o(n)}$, $L_F(m) = \frac{\beta^2}{2}(1 - Q)^2 + \frac{1}{n} \sum_i \log \frac{1}{1 - m_i^2}$ [1, Prop. 3.2, (3.5), p. 9], proved in §4 by free convolution. The gradient density uses $W_F m \stackrel{d}{=} Q^{1/2}v + \beta^{-1}\vartheta m$ with independent $v \sim N(0, I_n)$ and $\vartheta \sim N(0, \beta^2/n)$; $p_m(0)$ integrates over ϑ [1, Lemma 3.3, p. 9].
3. *Rate*. Sanov and Varadhan give

$$\limsup_n \frac{1}{n} \log \mathbb{E} \operatorname{Crit}_n \leq \sup_\rho \sup_{\vartheta} \{J(\rho, \vartheta) - H(\rho|\pi_0) - \vartheta^2/(2\beta^2) + \log 2\},$$

where J collects L_F and the density of Lemma 3.3 [1, Cor. 3.4, (3.6), pp. 9–10; Prop. 3.5, p. 10; App. B, pp. 32–35]. The supremum over ϑ is computed in closed form [1, §3.4, p. 11]:

$$\sup_{\vartheta} [\cdots] = \frac{1}{4\beta^2} \left(\beta^2(1 - q) + \frac{\beta\lambda\varphi^2}{q} - \frac{a_F}{q} \right)^2 - H(\rho|\rho_{\varphi,q}),$$

with (q, φ, a_F) the moments of ρ and $\rho_{\varphi,q}$ (density $h_{\varphi,q}$) the image of $N(\beta\lambda\varphi, \beta^2q)$ under $\tanh$.

4. *Bayes point* [1, Prop. 5.2, p. 22; proof p. 24]: the complexity $S_\star$ vanishes at $(q_\star, \varphi_\star, a_\star, e_\star)$, using $\xi \sim N(\lambda^2 q_\star, \lambda^2 q_\star)$, the identity $\mathbb{E} \tanh \xi = \mathbb{E} \tanh^2 \xi$ and $a_\star = \mathbb{E} \xi \tanh \xi$; for $\lambda \geq \lambda_0$, $S_\star \leq -c(\lambda)[(a - a_\star)^2 + (q - q_\star)^2 + (\varphi - \varphi_\star)^2 + (e - e_\star)^2]$ whenever $|q - q_\star|, |\varphi - \varphi_\star| \leq c_0/\lambda^2$ (a, e free).

CFM restrict the count to Hessian failures near the Bayes law, with g_n, H_n at $\beta = \lambda$ and a threshold t_H :

- *Localization* [2, Lemma 4.7, p. 17; proof pp. 41–43]: critical points with overlaps, entropy and free energy near the Bayes values lie in $\mathcal{D}_\eta$, where the empirical law of $\operatorname{arctanh} m_i$ is W_2 -within η of $N(\lambda^2 q_\star, \lambda^2 q_\star)$; the proof reduces to that product density and a W_2 deviation bound.
- *Restricted Kac–Rice* [2, Lemma 4.8, p. 17]: for $\mathcal{C} = \{m \in T : g_n(m) = 0, \ell(m, W) + U < t_H\}$ with $U \sim \operatorname{Unif}[-c, c]$, $\mathbb{E}|\mathcal{C}| \leq \int_T \mathbb{E}[|\det H_n(m)| \mathbf{1}\{\ell(m, W) + U < t_H\} \mid g_n(m) = 0] p_{g_n(m)}(0) dm$.
- *Conditional Hessian* [2, Lemma 4.9, p. 18; Lemma B.2, p. 43]: with $\ell_\varepsilon^+(m) = \inf\{\lambda_{\min} H_n(u) : \|u - m\| \leq \sqrt{\varepsilon n}\}$, every $m \in \mathcal{D}_\eta$ has $\mathbb{P}[\ell_\varepsilon^+(m) < 3t_H \mid g_n(m) = 0] \leq e^{-c_1 n}$ [2, proof of Cor. 4.10, p. 50]. The lower bound on its conditional mean comes from the comparison [2, Lemma 4.1, p. 15; proof App. A.4, p. 33]

$$\mathbb{E} \sup_x [v(x)^\top W_F v(x) + f(x)] \leq \mathbb{E} \sup_x \left[\frac{2}{\sqrt{n}} \|v(x)\| \langle \zeta, v(x) \rangle + f(x) \right], \quad \zeta \sim N(0, I_n),$$

with the TAP diagonal and Onsager term in f .

- *Assembly* [2, proof of Cor. 4.10, p. 50]: by Cauchy–Schwarz the restricted count is at most $\int_{\mathcal{D}_\eta} \mathbb{E}[|\det H_n|^2 \mid g_n = 0]^{1/2} p(0) dm \leq e^{c_1 n/4}$ (Lemma B.3, p. 49, from FMM's zero at the Bayes point) times $\sup_m \mathbb{P}[\ell_\varepsilon^+ < 3t_H \mid g_n = 0]^{1/2} \leq e^{-c_1 n/2}$.

Correspondence. On the Nishimori line $\lambda = \beta$, $j = \beta^2$, $W = \beta W_F$ and $\beta\lambda = j$; the preprint's y is $\operatorname{arctanh} m$, and its side-information field is $h_0 = t\mathbf{1} + \sqrt{t}g$.

Object	Preprint [10]	FMM / CFM
Disorder	$\tilde{J} = W + \frac{j}{n}\mathbf{1}\mathbf{1}^\top$ (5.4), p. 28	βY at $x = \mathbf{1}$ (FMM p. 8; CFM Lemmas 4.6–4.8)
Gradient, Hessian	$F_{\tilde{J}, h_0}(y) = y - h_0 + (jbI - \tilde{J})m$ (5.1), p. 26; $L = F'(y) = I + (jbI - W)V_y - \frac{j}{n}\mathbf{1}\mathbf{1}^\top V_y - \frac{2j}{n}mm^\top V_y$ (p. 38)	$F = g_n(\tanh y) - h_0$ and $L = H_n(m)V_y$ ((3.2)–(3.3))
Gradient law	$\operatorname{Cov}(Wm) = jqI + \frac{j}{n}mm^\top$; observation noise adds $(t + \sigma^2)I$ (proof of Lemma 5.3, p. 28)	$\operatorname{Cov}(\beta W_F m) = \beta^2(QI + mm^\top/n)$ (FMM Lemma 3.3)
Determinant	weight $e^{njb^2/2}$ in (5.5); Lemma 3.5; “the determinant rate in Lemma 3.5 matches the weight in (5.5)” (p. 38)	$e^{nL_F(m)}$ (FMM Prop. 3.2); first display below
Deterministic equivalent	$\mathbb{E}K_{ii} = 1 + O(n^{-1})$ (3.18), p. 16; $G_A \approx A$ (Lemma 3.4, (3.23), p. 19)	at spectral parameter 0 the Stieltjes fixed-point equation (4.4) has the root $1 - Q$, giving resolvent diagonal $1 - m_i^2$ (FMM (4.18), Lemma 4.9, p. 19)
Product density	$\prod_i \varphi_{d,s}(y_i)$, $d = t + jM$, $s = t + \sigma^2 + jq$ (5.6)–(5.7), p. 28	$h_{\varphi,q}$, i.e. $N(\beta\lambda\varphi, \beta^2q)$ in $\operatorname{arctanh}$ coordinates (FMM p. 11)
Rate	$\psi - \operatorname{KL}(P\ N(d, s))$; exponent identity in Lemma 5.3 (5.9) and p. 29	FMM §3.4 display; second display below
Large deviations	finite cover with Donsker–Varadhan tests against $N(d, s)^{\otimes n}$ (proof of Lemma 5.4, (5.10)–(5.11), pp. 30–31)	Varadhan upper bound and Sanov w.r.t. π_0 (FMM Prop. 3.5, App. B)
Bayes point	equality iff $P = N(d, s)$, where $a = sb$ (Lemma 4.1, (4.1), pp. 22–24); $\mathbb{E}_{N(s,s)} \tanh = \mathbb{E}_{N(s,s)} \tanh^2$ and $P_t = N(s_*, s_*)$ (4.10), Lemma 4.2, p. 25	$S_* = 0$ at $\xi \sim N(\lambda^2 q_*, \lambda^2 q_*)$, $\mathbb{E} \tanh \xi = \mathbb{E} \tanh^2 \xi$ (FMM Prop. 5.2)
Localization	weak neighborhood of P_t plus $\langle y^2 \mathbf{1}_{\{ y > R_1\}} \rangle \leq \nu$ (Lemma 5.4(ii),(iv), pp. 29–31)	W_2 -ball $\mathcal{D}_\eta$ of $N(\lambda^2 q_*, \lambda^2 q_*)$ in $\operatorname{arctanh} m$ (CFM Lemma 4.7)
Restricted count	$I_{t,\sigma}(E_t)$, with the conditional probability factor in (5.8)	CFM Lemma 4.8
Hessian failure	near P_t , conditional failure probability $\leq e^{-a_* n}$, simultaneously over $\ A - V_y\ _F \leq \epsilon_* \sqrt{n}$ (Lemma 5.5); conditional law (5.12) (p. 32)	$\mathbb{P}[\ell_\varepsilon^+(m) < 3t_H \mid g_n = 0] \leq e^{-c_1 n}$ on $\mathcal{D}_\eta$ (CFM Lemma 4.9; proof of Cor. 4.10, p. 50); conditional law CFM Lemma B.2

Object	Preprint [10]	FMM / CFM
Bulk comparison	$X_p = w^\top W w$ vs. $Y_p = 2\sqrt{j/n}\ w\ \zeta^\top w$, $w = A^{1/2}p$, shift $-\tau^2 j\chi_A\ w\ ^2$; increments differ by $\frac{2j}{n}[(\ w\ ^2 - \ w'\ ^2)^2 + 2(\ w\ \ w'\ - w^\top w')^2]$ (Lemma 3.2, p. 13); used in Lemma 5.5 (pp. 32, 34)	CFM Lemma 4.1 at GOE scale j ; App. A.4 bounds the same increment difference
Assembly	excluded scalar regions have negative rates; elsewhere $e^{-a_* n}$ times an unrestricted integral $\leq e^{\varepsilon n}$ (Lemma 5.4(i)) (5.25), p. 37	$e^{-c_1 n/2} \cdot e^{c_1 n/4}$ via Cauchy–Schwarz, Lemma B.3 (CFM p. 50)

Determinant weight. At $\lambda = \beta$ the Jacobian of F factors as $L = H_n(m)V_y$, and $dm = \prod_i (1 - m_i^2) dy$, so

$$\det L = \det H_n(m) \prod_i (1 - m_i^2), \quad e^{nL_F(m)} dm = e^{njb^2/2} \prod_i \frac{1}{1 - m_i^2} \cdot \prod_i (1 - m_i^2) dy = e^{njb^2/2} dy,$$

and Lemma 3.5's bound $\frac{1}{2n} \log \det(L^\top L + \gamma I) \leq \frac{jb^2}{2} + \frac{\varepsilon}{2}$ is FMM's $L_F(m)$ minus the Jacobian term. At $t = 0$ the center $z = y - jM\mathbf{1} + jbm$ of Lemma 5.3 is FMM's $u(m) = g_n(m) + \beta W_F m$, so as $\sigma \rightarrow 0$ the unrestricted integrand of Lemma 5.3 becomes FMM's bound $e^{nL_F(m)} p_m(0) dm$ on the Kac–Rice integrand, written in y .

Rate. The two closing computations are the same simplification, after FMM's supremum over ϑ and the preprint's exact Gaussian integral over the rank-one direction (prefactor $\sqrt{s/S}$):

$$\text{FMM, p. 11: } \frac{\beta^2(1-q)^2}{2} + \frac{\beta^2\chi_2^2}{4} + \chi_4 = \frac{1}{4\beta^2} \left(\beta^2(1-q) + \frac{\beta\lambda\varphi^2}{q} - \frac{a_F}{q} \right)^2,$$

$$\text{Preprint, p. 29: } -\frac{Ba}{s} - \frac{B^2q}{2s} + \frac{j(a+Bq)^2}{2sS} + \frac{jb^2}{2} = \frac{j(a-sb)^2}{2sS} \quad (B = jb),$$

where χ_2, χ_4 are FMM's coefficients and $\beta^2(1-q)^2/2$ and $jb^2/2$ are the determinant terms. At $t = \sigma = 0$ and $\lambda = \beta$: $\varphi = M$, $d = jM = \beta\lambda\varphi$, $s = jq$, $S = 2jq$ and $a = a_F - jM^2$, so

$$\psi = \frac{j(a-sb)^2}{2sS} = \frac{(a_F - j\varphi^2 - jq(1-q))^2}{4jq^2} = \frac{1}{4\beta^2} \left(\beta^2(1-q) + \frac{\beta\lambda\varphi^2}{q} - \frac{a_F}{q} \right)^2,$$

and $\text{KL}(P\|N(jM, jq)) = H(\rho|\rho_{\varphi,q})$ because relative entropy is invariant under the bijection $\tanh$ (ρ is the law of $\tanh y$ under P). The preprint's rate is this functional with the reference Gaussian shifted by t in mean and $t + \sigma^2$ in variance (so $S = s + jq$ replaces $2jq$).

What is new in the preprint.

- *Side information below the threshold.* FMM and CFM have no external field. The field $h_0 = t\mathbf{1} + \sqrt{t}g$ moves the Bayes law to $N(s_*(t), s_*(t))$ with $s_*(t) > 0$ for $t > 0$, so the count is nontrivial for $j < 1$, where without a field $q_* = 0$ [1, Prop. 5.5, p. 23].
- *A global sharp sign.* Lemma 4.1 gives $\text{KL}(P\|N(d, s)) \geq j(a-sb)^2/(2sS)$ for every P , every $s \geq jq$ and every $0 < j < 1$, with equality only at $N(d, s)$, by the monotone transport $y \mapsto y - u \tanh y$ and a Gaussian second-moment comparison. FMM's negativity is local and needs $\lambda \geq \lambda_0$ large; CFM localize first instead of proving a sign.

- *The determinant rate re-proved* (Lemma 3.5) by integrating loop equations in the coupling $\tau \in [0, 1]$, with γ -regularization, simultaneously over all diagonals V and finite-rank E , in place of FMM's free-convolution route at fixed m .
- *Conditional-Hessian positivity re-proved* (Lemma 5.5): whitening by S_A , Gram entries from Lemma 3.4, and an explicit 3×3 Gram matrix with Schur-complement determinant $(d_0 + 2g_1)(d_0 + g_1)^2 > 0$ (5.16)–(5.17), pp. 33–34, in place of CFM's Sudakov–Fernique variational formula; Lemma 3.2 itself is unconditional, along $\tau \in [0, 1]$, with a one-line scalar reduction.
- *Perturbed diagonals*: positivity over a Frobenius ball of diagonals A around V_y , where CFM use a ball of points u .
- *Tail localization by an exponential tilt* (Lemma 5.4(iv)) in place of CFM's W_2 deviation bound; no Cauchy–Schwarz, as the determinant enters as a deterministic weight.
- *Uniformity in $t \in [0, T]$* : uniform rates, a small-time margin (5.23), p. 37 and a time mesh (pp. 38–39).

How to check.

1. *Rate.* (5.6)–(5.7), p. 28, the exponent identity in the proof of Lemma 5.3, p. 29 and (5.9), p. 29, against FMM Lemma 3.3 (p. 9) and the §3.4 display (p. 11): substitute $s = jq$, $S = 2jq$, $a = a_F - j\varphi^2$ as above.
2. *Weight.* (5.5), p. 28, the display of L and the remark on the determinant rate (p. 38), and Lemma 3.5, p. 20, against FMM (3.3) (p. 8) and Prop. 3.2 (p. 9): check $L = H_n(m)V_y$ and the Jacobian identity.
3. *Bayes point.* The equality case of Lemma 4.1 (p. 24) and (4.10) in the proof of Lemma 4.2 (p. 25), against FMM Prop. 5.2 (p. 22) and its proof (p. 24).
4. *Restricted count.* E_t and (5.25), p. 37, Lemma 5.5, p. 31 and Lemma 3.2, pp. 12–13, against CFM Lemmas 4.7–4.8 (p. 17), Lemma 4.9 and Cor. 4.10 (p. 18), the proof of Cor. 4.10 (p. 50), and Lemma 4.1 (p. 15) with App. A.4 (p. 33).

3.2 S2. Approximate-critical-point control from annealed Kac–Rice bounds

Summary. To exclude unstable *approximate* solutions of the TAP field equation, the preprint uses the device of Huang and Sellke (HS) [3] that turns an exponentially small annealed Kac–Rice bound on exact critical points with a bad property into high-probability non-existence of approximate ones [3, §5: Theorem 5.2, p. 28; Lemma 5.11, pp. 33–35; proof of Theorem 5.2, p. 36]. The device has four parts: a Gaussian rerandomization, so that approximate critical points contribute to a Kac–Rice integral; a local lower bound in which the inverse-Jacobian volume of a region adapted to the Jacobian cancels the determinant factor; inheritance of the bad property on that region; and Markov's inequality against the annealed bound. In the preprint these are the smoothed integral (5.5) with the convolution in the proof of Lemma 5.3 (p. 28), Lemma 5.6 (pp. 35–37), and the exclusion step in the proof of Proposition 5.1 (pp. 37–38). The preprint does not cite HS and presents the step as its own (§1.1, p. 4; §1.3, p. 5); Sellke appears in its bibliography only as a coauthor of El Alaoui–Montanari–Sellke [5], cited for Gaussian observations, planted laws and AMP, which contains no such step. Grade: STRONG, because all four parts reappear for HS's purpose: ruling out approximate critical points with an atypical Hessian.

Setup. Section 2 defines $F = F_{\tilde{J}, h_0}$, $H_J(y, A)$, the planted law (5.4) and $I_{t, \sigma}$ (5.5). This finding also needs:

- *Jacobian.* $L = F'(y) = I + (jbI - \tilde{J})V_y - \frac{2j}{n}mm^\top V_y$ (not symmetric) and $Q = L^\top L + \gamma I$, $\gamma > 0$ fixed (Lemma 5.6).
- *Bad set and annealed bound.* For $t \geq t_0$, $E_t(W)$ is the set of y for which some diagonal $0 \leq A \leq (1 + \eta)I$ with $\|A - V_y\|_F \leq \epsilon_* \sqrt{n}$ has $\lambda_{\min}(\mathbf{H}_{\tilde{J}}(y, A)) \leq c_E$; for $t < t_0$, $E_t = \{q(y) \geq q_0\}$ (p. 37). The Kac–Rice computation of finding S1 gives $\mathbb{E}_{I_{t,\sigma}}(E_t) \leq e^{-a_0 n}$ for $0 \leq t \leq T$, $0 < \sigma \leq \sigma_*$ (5.25).
- *Witness* (5.26): $\|F(y)\| \leq \rho \sqrt{n}$, $\|A - V_y\|_F \leq \frac{1}{2} \epsilon_* \sqrt{n}$ and $\lambda_{\min}(\mathbf{H}_{\tilde{J}}(y, A)) \leq \frac{1}{2} c_E$.
- *Determinant event* (Lemma 3.5, p. 20, applied on p. 38): with probability at least $1 - Ce^{-a_1 n}$, simultaneously for all y , $n^{-1} \log \det Q \leq jb(y)^2 + \varepsilon_1$.
- *Reading of (5.5).* For a standard Gaussian $\tilde{g}$, $F_{\tilde{J}, h_0 + \sigma \tilde{g}} = F_{\tilde{J}, h_0} - \sigma \tilde{g}$, so the area formula gives

$$\mathbb{E}_{\tilde{g}} \#\{y \in E : F_{\tilde{J}, h_0 + \sigma \tilde{g}}(y) = 0\} = (2\pi\sigma^2)^{-n/2} \int_E e^{-\|F(y)\|^2/(2\sigma^2)} |\det L(y)| dy.$$

So $I_{t,\sigma}(E)$ is the right-hand side with $|\det L(y)|$ replaced by the weight $e^{njb(y)^2/2}$, calibrated to its exponential scale (p. 28): the smoothed, weighted count $\mathbb{E}_{\tilde{g}} \sum e^{njb(y)^2/2} / |\det L(y)|$ over the zeros $y \in E$ of the rerandomized $F_{\tilde{J}, h_0 + \sigma \tilde{g}}$. In the proof of Lemma 5.3 the rerandomization turns $\sqrt{t}g$ into $\sqrt{t + \sigma^2}g'$.

Role in the argument. Proposition 5.1 must hold at every approximate zero ($\|F\| \leq 2\rho_0 \sqrt{n}$), not only at exact zeros: Lemma 5.7 (p. 40) turns positivity on the whole small-residual region into a unique zero, within $C\|F(y)\|$ of every y there, and §§6–7 compare the approximate fields of the recursion (6.1) with that zero (p. 45). The device turns (5.25) into the statement that at a fixed time no witness (5.26) exists, except with probability $Ce^{-a_2 n}$; diagonal removal, a time mesh and the planted-to-ordinary transfer then give Proposition 5.1 (pp. 38–40).

The earlier technique. HS study a multi-species spherical spin glass H_N on a product of spheres S_N , with Riemannian gradient and Hessian $\nabla_{\text{sp}}, \nabla_{\text{sp}}^2$, rescaled radial derivative ∇_{rad} , and distorted Hessian $J_N(x)$ ((5.11), p. 32). $\text{Crt}_N^{\overline{\mathcal{D}}, \varepsilon, v}(H_N)$ is the set of x with $\|\nabla_{\text{sp}} H_N(x)\| \leq \varepsilon \sqrt{N}$ whose tuple (radial derivative, overlap with the external field, energy, Hessian spectrum, spectral law) is v -far from a set $\overline{\mathcal{D}}$ ((5.2), p. 28).

- *Theorem 5.2* (p. 28), introduced as “of significant independent interest”: if $\mathbb{E}|\text{Crt}_N^{\overline{\mathcal{D}}, 0, v/2}(H_N)| \leq e^{-c_0 N}$ (5.3), then for some small $\varepsilon = \varepsilon(\xi, v, c_0)$, $\mathbb{P}[\|\text{Crt}_N^{\overline{\mathcal{D}}, \varepsilon, v}(H_N)\| \geq 1] \leq e^{-cN}$.
- *Ingredients.* The rerandomization $H_{N,\delta} = \sqrt{1 - \delta} H_N + \sqrt{\delta} H'_N$ with H'_N an independent copy (5.1); the constants $(\varepsilon, \alpha, \eta, \iota) = (\delta^{10}, \delta^{1/3}, \delta^{1/10}, \delta^{1/100})$ ((5.12), p. 32); and a typical set $K_N(\varepsilon)$ of probability at least $1 - e^{-c(\varepsilon)N}$ on which, simultaneously for all x , the spectral law of $J_N(x)$ is ε -close to its deterministic counterpart (Definition 13 and Proposition 5.8, p. 32).
- *Lemma 5.11* (pp. 33–35). If $H_N \in K_N(\varepsilon)$ and $\|\nabla_{\text{sp}} H_N(x)\| \leq \varepsilon \sqrt{N}$ (5.15), then in expectation over H'_N there are at least $e^{-o_\delta(N)}$ critical points $y \in B_{\alpha \sqrt{N}}(x)$ of $H_{N,\delta}$ whose energy, radial derivative and Hessian are within $\iota^2 N$, ι^2 , ι of those of H_N at x ((5.16)–(5.19)). Proof: the Kac–Rice lower bound (5.20) is the density at 0 of $\nabla_{\text{sp}} H_{N,\delta}(y)$ given H_N (5.22) times the conditional expected $|\det \nabla_{\text{sp}}^2 H_{N,\delta}(y)|$, which Lemma 5.9 (p. 32) evaluates as $\exp(N \int \log |\lambda| d\mu_{\xi, \nabla_{\text{rad}} H_N(x)} \pm o_\iota(N))$ (5.21). The density is integrated over balls S_1 of radius $\alpha^2 \sqrt{N/2}$ in the span $T_1(x)$ of eigenvectors of $J_N(x)$ with eigenvalues in $[-\eta, \eta]$ and S_2 of radius $\alpha \sqrt{N/2}$ in its complement, with the Taylor step Lemma 5.10. The change of variables

$v_2 = J_N(x)s_2$ gives $|\det(J_N(x)|_{S_2(x)})|^{-1}$ (5.23), which by $K_N(\varepsilon)$ cancels (5.21) up to $e^{o_\delta(N)}$ (p. 35).

- *Proof of Theorem 5.2* (p. 36). By (5.16)–(5.19), Proposition 2.4 and $y \in B_{\alpha\sqrt{N}}(x)$ (for the overlap), these critical points are $v/2$ -far from $\overline{\mathcal{D}}$, so their expected number given H_N is at least $e^{-o_\varepsilon(N)} \mathbf{1}\{H_N \in K_N(\varepsilon, v)\}$ (5.24), where $K_N(\varepsilon, v)$ is the set of $H_N \in K_N(\varepsilon)$ with $\text{Crt}_N^{\overline{\mathcal{D}}, \varepsilon, v}(H_N) \neq \emptyset$. Since $H_{N, \delta}$ has the law of H_N , averaging against (5.3) gives $\mathbb{P}[H_N \in K_N(\varepsilon, v)] \leq e^{-cN}$ once ε is small given c_0 .

Correspondence. Dictionary: residual tolerance $\rho \leftrightarrow \varepsilon$, smoothing variance $\sigma^2 \leftrightarrow \delta$, radius $r \leftrightarrow \alpha$, regularization $\sqrt{\gamma} \leftrightarrow \eta$, annealed rate $a_0 \leftrightarrow c_0$, the preprint’s local-volume loss ε ((5.20)) $\leftrightarrow o_\delta(1)$; F and L play $\nabla_{\text{sp}} H_N$ and J_N . (So HS’s ε is not the preprint’s; HS’s r in $N - r$ is the number of species.)

Step	Preprint [10]	HS [3]
Annealed input	$\mathbb{E} I_{t, \sigma}(E_t) \leq e^{-a_0 n}$ (5.25), p. 37	$\mathbb{E} \text{Crt}_N^{\overline{\mathcal{D}}, 0, v/2} \leq e^{-c_0 N}$ ((5.3), p. 28)
Bad point	witness (5.26)	$x \in \text{Crt}_N^{\overline{\mathcal{D}}, \varepsilon, v}$ ((5.2), (5.15))
Smoothing	field rerandomized by $\sigma \tilde{g}$; kernel of variance σ^2 at $F(y)$ in (5.5) (p. 28)	$H_{N, \delta}$ (5.1); kernel of variance $\delta/(1 - \delta)$ at $\nabla_{\text{sp}} H_N(y)$ ((5.20), (5.22))
Det. weight	deterministic $e^{njb(y)^2/2}$ (5.5)	conditional $\mathbb{E} \det \nabla_{\text{sp}}^2 H_{N, \delta}(y) $ (Lemma 5.9, (5.21))
Det. at the point	Lemma 3.5 event, all y at once (p. 38)	$K_N(\varepsilon)$, all x at once (Def. 13(c), Prop. 5.8; used p. 35)
Region and volume	ellipsoid $y + \sigma Q^{-1/2} g''$, $\ g''\ \leq 2\sqrt{n}$, inside the $r\sqrt{n}$ -ball if $2\sigma/\sqrt{\gamma} \leq r$; factor $(\det Q)^{-1/2}$ (5.21), p. 36	$S_1 \oplus S_2$, capturing half the Gaussian mass as $\alpha\eta \gg \sqrt{\delta}$; factor $ \det(J_N(x) _{S_2(x)}) ^{-1}$ ((5.23), pp. 34–35)
Taylor step	$R_F = F(y + \Delta) - F(y) - L\Delta$ (5.22), p. 36	Lemma 5.10 ((5.14), p. 33)
Cancellation	(5.19) cancels $jb(y)^2/2$ up to $\varepsilon_1/2$, giving $\geq e^{-\varepsilon n}$ (5.20) (p. 37)	$ \det ^{-1}$ cancels (5.21), leaving $e^{-o_\delta(N)}$ (p. 35)
Inheritance	the same A witnesses instability on the $r\sqrt{n}$ -ball, so the ball lies in E_t (for $t < t_0$, Lipschitz continuity of q) (pp. 37–38)	(5.16)–(5.19), Prop. 2.4 and $y \in B_{\alpha\sqrt{N}}(x)$ make the nearby critical points $v/2$ -far (p. 36)
Markov	witness on the determinant event $\Rightarrow I_{t, \sigma}(E_t) \geq e^{-\varepsilon n}$; Markov and (5.25) give $Ce^{-a_2 n}$ (p. 38)	(5.24) averaged against (5.3): $e^{-c_0 N} \geq e^{-o_\varepsilon(N)} \mathbb{P}[K_N(\varepsilon, v)]$ (p. 36)
Constants	$\varepsilon < a_0/2$, then ε_1 , then γ , then σ , then ρ (Lemma 5.6, pp. 35–36; p. 38)	(5.12), then ε small given c_0 (pp. 32, 36)

As the device requires, residual $\ll$ smoothing $\ll$ neighborhood: $\rho \ll \sigma \leq r\sqrt{\gamma}/2$ against $\varepsilon = \delta^{10} \ll \sqrt{\delta} \ll \alpha\eta$. The local lower bounds (HS (5.23); the preprint’s (5.21)) and the determinant inputs

that cancel them read, with HS's species factors omitted:

$$\begin{aligned}
(2\pi\delta)^{-\frac{N-r}{2}} \int_{B_{\alpha\sqrt{N}}(x)} e^{-\frac{1-\delta}{2\delta} \|(\xi')^{-1/2} \diamond \nabla_{\text{sp}} H_N(y)\|^2} dy &\geq \frac{e^{-\iota N} \text{Vol}(S_1(x))}{2(1+\iota)^{\frac{N-r}{2}}} |\det(J_N(x)|_{S_2(x)})|^{-1}, \\
(2\pi\sigma^2)^{-\frac{n}{2}} \int_{\|y'-y\| \leq r\sqrt{n}} e^{-\frac{\|F(y')\|^2}{2\sigma^2} + \frac{nb(y')^2}{2}} dy' &\geq (\det Q)^{-\frac{1}{2}} \mathbb{P}(\mathcal{B}) \mathbb{E} \left[e^{\frac{\|g''\|^2}{2} - \frac{\|F(y+\Delta)\|^2}{2\sigma^2} + \frac{nb(y+\Delta)^2}{2}} \mid \mathcal{B} \right]; \\
\det(J_N(x)|_{S_2(x)}) &= e^N \int \log |\lambda| d\mu_{\xi, \nabla_{\text{rad}} H_N(x) \pm o_\delta(N)} \quad \text{vs.} \quad \frac{1}{n} \log \det Q \leq nb(y)^2 + \varepsilon_1.
\end{aligned}$$

What is new in the preprint.

- Smoothing acts on the field only. Since h_0 does not enter $H_J(y, A)$ (p. 27), inheritance is a deterministic Lipschitz bound, and HS's control of the rerandomized Hessian (Lemma 5.9 and its use on p. 33) has no counterpart.
- No exact zeros of a rerandomized system are counted: the smoothed integral, with its deterministic weight, is itself the annealed object; Lemma 5.3 reduces it to a scalar integral against a conditional probability; Lemma 5.4 bounds the former and Lemma 5.5 the latter, uniformly in $0 < \sigma \leq \sigma_*$, giving (5.25).
- Regularization $Q = L^\top L + \gamma I$ of a non-symmetric Jacobian, a Gaussian ellipsoid and Jensen's inequality replace the near-kernel split $S_1 \oplus S_2$, which uses the eigenvectors of a symmetric Hessian.
- The Ising TAP field equation, a bad set defined through diagonals A near V_y , and uniformity in $t \in [0, T]$.

How to check.

1. *Statements.* HS Theorem 5.2 and the plan opening HS §5 (p. 28) against the opening of §5.3 (p. 35) and the exclusion paragraph in the proof of Proposition 5.1 (pp. 37–38).
2. *Smoothing.* HS (5.1), (5.20), (5.22) (pp. 28, 33–34) against (5.5) and the proof of Lemma 5.3 (p. 28), via the area formula in the Setup.
3. *Local lower bound.* The proof of HS Lemma 5.11 from (5.22) to its end (pp. 34–35), with Lemma 5.10 (p. 33), against the proof of Lemma 5.6 (pp. 36–37): region, change of variables, Taylor remainder, cancellation.
4. *Inheritance, Markov, constants.* HS's proof of Theorem 5.2 (p. 36) and (5.12) (p. 32) against p. 38: the ball inside E_t , the choice of $\varepsilon, \varepsilon_1, \gamma, \sigma, \rho$, and Markov with (5.25).

4 Partial findings

4.1 P1. The Stein equation for tanh under a cosh-tilted Gaussian, used to compare a field with an Onsager-corrected field

Summary. Chatterjee [4] derives the TAP equations by Stein's method for the cosh-tilted Gaussian laws $\psi \propto \cosh(ax + b)e^{-(x-\mu)^2/2\sigma^2}$: a Stein operator whose equation for $u(x) = \tanh(ax + b)$ has right side $\tanh(ax + b) - \tanh(a\mu + b)$, its bounded solution, and its use at the Gibbs local field, centred at the Onsager-corrected field, with the Onsager coefficient as variance parameter (in the units $z = \beta\ell + h$) [4, §1.8, (12)–(15), p. 7; Lemma 1.3, p. 8; §3.4, pp. 20–23; Corollary 1.2, p. 5]. The preprint's scalar identity (6.8)–(6.9), p. 45 is this Stein equation, with the solution in closed form. It drives the induction step of the weak residual rule (proof of Lemma 6.5, p. 54), where at $p = 1$ it compares the Gibbs-sample local field with an Onsager-corrected field, and it

gives the comparison with the TAP root behind Proposition 7.2 ((6.12)–(6.13), pp. 45–46; (7.3), p. 57). The preprint does not cite Chatterjee; it introduces (6.9) as “a scalar identity valid for arbitrary real fields” and cites Bolthausen [12] only for the Onsager-corrected form of the recursion (§1.1, p. 4; §6.1, p. 42). Grade: PARTIAL, because the operator, its bounded solution and the field–centre–parameter configuration are reused; Chatterjee’s L^2 estimate of $\langle Tf \rangle$, the main work of his method, is not.

Setup. Section 2 defines $J, j = \beta^2, \mu_{h_0}$, the recursion (6.1) $(m^l, b_l, h^l, R_l), F_{J, h_0}, d_i$ and $\|\cdot\|_*$. In (6.1) the field h^{l+1} carries the Onsager term $-jb_l m^{l-1}$; $h^1 = h_0 + Jx$ carries none. This finding also needs:

- *Kernel* (6.8): for $z, r_0, a \in \mathbb{R}$, $\Phi(z; r_0, a) = \int_0^1 e^{a(1-u^2)/2} \cosh(u(z - r_0)) du / (\cosh z \cosh r_0)$.
- *Tests* (Definition 6.2, p. 44): vectors U built from the primary fields by finite recipes. The site partial ∂_l differentiates the explicit argument h_i^l with all else fixed; “admissible from level $p+1$ ” means every explicit primary argument has index at least $p+1$, so $\partial_p U = 0$. Auxiliary fields carry an Onsager correction, $y = Jw - j \sum_l n^{-1} (\sum_i \partial_l w_i) m^{l-1} - \dots$ (6.7).
- *TAP root*: at a stable field, r is the zero of F_{J, h_0} , $t_* = \tanh r$, $q_* = n^{-1} \|t_*\|^2$, so $r = h_0 + Jt_* - j(1 - q_*)t_*$ (§6.4, p. 45).
- *Conditional-spin identity*: since $\mathbb{E}_{\mu_{h_0}}[x_i | x_{-i}] = m_i^1 = \tanh h_i^1$ and $v_i^1 = 1 - (m_i^1)^2$, $\mathbb{E} \sum_i (x_i - m_i^1) G U_i = \mathbb{E} \sum_i v_i^1 d_i(G U_i)$ (6.31), p. 53.

Role in the argument. Lemma 6.5 bounds tested residuals, $|\mathbb{E} G R_p^\top U| \leq C \|G\|_*$, with a square-reweighted version (6.30). It yields the moment bounds on $S_l = R_l^\top m^l / \sqrt{n}$ that select two small consecutive residuals (Lemma 7.1) and the residual estimates in Propositions 7.2 and 7.3, i.e. input 2 of Section 2. Its base case is (6.31); its induction step applies (6.9) at $(z, r_0, a) = (h_i^p, h_i^{p+1}, j(b_p - b_{p-1}))$ to rewrite $R_{p+1} = \tanh h^p - \tanh h^{p+1}$ through earlier residuals. In the proof of Proposition 7.2, (6.9) at $(h_i^k, r_i, j(1 - b_{k-1} - q_*))$ and its r_0 -derivative turn the implicit test vector into the magnetization error $(m^k - t_*)^\top e$ and the coefficient discrepancy L_* ; this gives the directional covariance bound (7.2) (pp. 56–58).

The earlier technique. Chatterjee’s Gibbs measure is $G_N(\sigma) \propto \exp\{\beta N^{-1/2} \sum_{i < j} g_{ij} \sigma_i \sigma_j + h \sum_i \sigma_i\}$, and $\langle \cdot \rangle$ there is the Gibbs average.

- *Fields* [4, (5), p. 3; (10), p. 5]: the local field $\ell_i = N^{-1/2} \sum_{j \neq i} g_{ij} \sigma_j$ and the Onsager-corrected centre $r_i = N^{-1/2} \sum_{j \neq i} g_{ij} \langle \sigma_j \rangle - \beta(1 - q) \langle \sigma_i \rangle$, so that $\beta r_i + h$ is the argument of $\tanh$ in the TAP equations.
- *Stein’s method for mixture Gaussians* [4, §1.8, p. 7]: $M(a, b, \mu, \sigma^2)$ has density $\psi \propto \cosh(ax + b) e^{-(x-\mu)^2/2\sigma^2}$ (12), a mixture of $N(\mu \pm a\sigma^2, \sigma^2)$; $\int \tanh(ax + b) \psi = \tanh(a\mu + b)$ (14); the operator $Tf = f' - (\frac{x-\mu}{\sigma^2} - a \tanh(ax + b))f$ characterizes M ; and the Stein equation is $Tf = u - \int u \psi$ (15). Lemma 1.3 (p. 8) gives a solution $f(x, \mu)$ with $|f|, |\partial_x f|, |\partial_\mu f| \leq C \|u\|_\infty$ uniformly in (x, μ) , namely $f = \rho^{-1} \int_{-\infty}^x \rho(t, \mu) (u(t) - \int u \psi) dt$ with $\rho = \cosh(ax + b) e^{-(x-\mu)^2/2\sigma^2}$ (§3.6, p. 25). Chatterjee identifies the correction term in the mixture identity (9) with the Onsager correction (pp. 4–5).
- *Application* [4, Theorem 1.1, p. 5; §3.4, pp. 20–23]: the law of ℓ_1 is close to $M(\beta, h, r_1, 1 - q)$. With f solving (15) for these parameters ((38), p. 20), Gaussian integration by parts in $(g_{1j})_j$ (Lemma 1.4, p. 9) and the overlap concentration (2) (p. 2), followed by the standard conditional-mean identity $\langle \sigma_1 f \rangle = \langle \tanh(\beta \ell_1 + h) f \rangle$ (p. 22), show that $\langle (\ell_1 - r_1 - \beta(1 - q) \tanh(\beta \ell_1 + h)) f \rangle -$

$(1-q)\langle\partial_x f\rangle = -(1-q)\langle Tf\rangle$ is small in L^2 over the disorder. With $u = \tanh(\beta x + h)$ and (14), this is Corollary 1.2 (p. 5): $\langle\sigma_i\rangle \approx \tanh(\beta r_i + h)$.

The substitution $z = \beta x + h$ maps $M(\beta, h, \mu', 1-q)$ to $M(1, 0, \beta\mu' + h, \beta^2(1-q))$. In these units the centre is the TAP field $\beta r_i + h$ and $\sigma^2 = \beta^2(1-q)$ is its Onsager coefficient, and with $(a, b) = (1, 0)$,

$$-\sigma^2 Tf = (x - \mu - \sigma^2 \tanh x)f - \sigma^2 \partial_x f$$

for every f , and the Stein equation (15) for $u = \tanh$ reads $Tf = \tanh x - \tanh \mu$.

Correspondence. In the table below Chatterjee's tilt is $(1, 0)$, and his μ and σ^2 are the preprint's r_0 and a ; rows 6–8 are the $p = 1$ step of Lemma 6.5.

Object	Preprint [10]	Chatterjee [4]
Operator	$(z - r_0 - a \tanh z) - a \partial_z$ (6.9)	$-\sigma^2 T$, $T = \partial_x - \left(\frac{x-\mu}{\sigma^2} - \tanh x\right)$ (§1.8)
Right side	$\tanh z - \tanh r_0$ (6.9)	$\tanh x - \int \tanh \psi = \tanh x - \tanh \mu$ ((14), (15))
Solution	$\Phi(z; r_0, a)$ (6.8), any $a \in \mathbb{R}$	$f(x, \mu)$ (§3.6); $\Phi = -f/\sigma^2$ for $\sigma^2 = a > 0$
Bounds	$0 < \Phi \leq e^{ a /2}$; for a in a bounded interval, each fixed-order derivative in (z, r_0, a) is at most $C\Phi$, uniformly in z, r_0 (p. 45)	$ f , \partial_x f , \partial_\mu f \leq C\ u\ _\infty$, uniformly in x, μ (Lemma 1.3)
Centre derivative	$[(z - r_0 - a \tanh z) - a \partial_z] \partial_{r_0} \Phi = \Phi - \text{sech}^2 r_0$ (p. 46)	f defined on $\mathbb{R}^2$; $\partial_\mu f$ enters at $\mu = r_1$ (§3.4, (39), pp. 21–22)
Field z	$h_i^1 = h_{0,i} + (Jx)_i$, at the Gibbs sample x	$\beta \ell_1 + h$, at a Gibbs sample (5)
Centre r_0	$h_i^2 = h_{0,i} + (Jm^1)_i - j b_1 x_i$	$\beta r_1 + h = h + \beta N^{-1/2} \sum_j g_{1j} \langle \sigma_j \rangle - \beta^2(1-q) \langle \sigma_1 \rangle$ (10)
Parameter	$a_1 = j b_1$, the Onsager coefficient of h^2 (h^1 has none); generally the centre's minus the field's Onsager coefficient, $j(b_p - b_{p-1})$ or $j(1 - b_{k-1} - q_*)$	$\sigma^2 = \beta^2(1-q)$, the Onsager coefficient of $\beta r_1 + h$ ($\beta \ell_1 + h$ has none)
TAP comparison	$(m^k - t_*)^\top e = \sum_i [(h_i^k - r_i - a m_i^k) - a \partial_k] (\Phi(h_i^k; r_i, a) e_i)$, with r the TAP root (6.13), p. 46; (7.3), p. 57	$\langle \tanh(\beta \ell_i + h) \rangle$ against $\tanh(\beta r_i + h)$ through T with $u = \tanh(\beta x + h)$ (Corollary 1.2)

At $p = 1$ the preprint's centre and parameter have Chatterjee's form with the Gibbs means $\langle \sigma_j \rangle$ replaced by the conditional means m_j^1 at the sample, $\langle \sigma_1 \rangle$ by x_1 , and $1 - q \approx N^{-1} \sum_i (1 - \langle \sigma_i \rangle^2)$ by $b_1 = n^{-1} \sum_i (1 - (m_i^1)^2)$; the values differ. Side by side, with $z = \beta \ell_1 + h$, $\mu = \beta r_1 + h$, $\sigma^2 = \beta^2(1-q)$ on the left [4, p. 22] and $U'_i = \Phi(h_i^p; h_i^{p+1}, a_p) U_i$, $\partial_p U = 0$ on the right (proof of Lemma 6.5, p. 54):

$$\begin{aligned} \langle (\ell_1 - r_1 - \beta(1-q) \tanh(\beta \ell_1 + h)) f \rangle - (1-q) \langle \partial_x f \rangle &= \beta^{-1} \langle (z - \mu - \sigma^2 \tanh z) f - \sigma^2 \partial_z f \rangle, \\ R_{p+1}^\top U &= \sum_i [(h_i^p - h_i^{p+1} - a_p \tanh h_i^p) - a_p \partial_p] U'_i = (h^p - h^{p+1} - a_p m^p)^\top U' - a_p \sum_i \partial_p U'_i. \end{aligned}$$

What is new in the preprint.

- The identity is used exactly and pointwise at fixed J , with no claim about the law of any field. It holds for every real a , and is applied with $a_p = j(b_p - b_{p-1})$ ($|a_p| \leq j$; $a_1 = jb_1 \geq 0$, either sign for $p \geq 2$) and with $a = j(1 - b_{k-1} - q_*)$, which is $O(\rho)$ by (6.11) and of either sign.
- The closed form gives, for a in a bounded interval, bounds on derivatives of every order in (z, r_0, a) and on the ratios $\partial^\alpha \Phi / \Phi$, uniformly in the fields, so that ΦU can enter further recipes; the r_0 -derivative identity yields the discrepancy L_* .
- Chatterjee's L^2 estimate of $\langle Tf \rangle$ (Gaussian integration by parts over the disorder and overlap concentration) is replaced by algebra: $h^p - h^{p+1} - a_p m^p = JR_p + a_p R_p - jb_{p-1} R_{p-1}$, symmetry of J with the Onsager-corrected auxiliary field (6.7), cancellation of $-a_p \sum_i \partial_p U'_i$ against the Onsager term through (6.34), and strong induction on p down to (6.31), including the square-reweighted form (6.30) (pp. 53–54).
- Centres built from one sample (m^1, x, b_1) rather than Gibbs averages; validity for all $\beta < 1$ at stable fields rather than under the overlap hypothesis (2), which Chatterjee has for $\beta \leq \beta_0$.

How to check.

1. *The identity.* (6.8)–(6.9), p. 45 against Chatterjee §1.8 (operator T , (14), (15)), p. 7, and the formula for f in §3.6, p. 25: set Chatterjee's tilt to $(1, 0)$, his μ to r_0 and his σ^2 to the preprint's a ; then $-\sigma^2 T$ is the operator of (6.9), and $\Phi = -f/\sigma^2$ (both are bounded solutions; the homogeneous solution $1/\rho$ is unbounded).
2. *The $p = 1$ configuration.* The induction step in the proof of Lemma 6.5, p. 54 with (6.1) at $p = 1$, against Chatterjee (5), (10), pp. 3–5, and the display on p. 22, after $z = \beta \ell_1 + h$.
3. *The TAP-root comparison.* (6.12)–(6.13), pp. 45–46 and (7.3) with the paragraph after it, p. 57 against Corollary 1.2, p. 5, and the remark on p. 11 that the TAP argument is $\beta r_i + h$.

4.2 P2. The planted stochastic-localization framework for SK at its points of use: planted law with likelihood transfer, and the state-evolution fixed point

Summary. Proposition 5.1 is proved under a planted law and then transferred to the SK disorder. That law, with the likelihood identity behind the transfer, is the planted model of El Alaoui, Montanari and Sellke (AMS) for the SK observation process [5, §4.1, (4.1)–(4.9), Proposition 4.2, (4.14), pp. 10–12], and the equality law that locates the zero set of the preprint's Kac–Rice rate is the AMS state-evolution fixed point, in the form given by their Nishimori identity [5, (2.5), (4.16), Proposition 4.4, Lemma 4.5(c), pp. 5, 12–14]. In the preprint they are the planted law (5.4), p. 28, its derivation and transfer (§5.4, pp. 39–40), and Lemma 4.2 (pp. 25–26); the Kac–Rice computation run under this law is finding S1. AMS is cited once, in the introduction (§1.1, p. 4), as a parallel combination of Gaussian observations, planted laws and AMP; nothing is cited where these ingredients are used. Grade: PARTIAL, because credit for the planted law is present only at claim level and as a parallel use, and the fixed point is a fragment of the AMS state-evolution lemma, re-proved largely by a different route.

Setup. Section 2 defines J , μ_h , $Y(t)$, the planted law (5.4), the integral (5.5), its rate and s_* . This finding also needs:

- *Size bias.* Under the original law, J is the zero-diagonal disorder, $X \sim \mu_0$ given J , and $Y(t) = tX + B(t)$ (Proposition 5.1, p. 27). Put $Z(J) = \sum_x e^{x^\top Jx/2}$, $L(J) = Z(J)/\mathbb{E}Z(J)$ and

$D_X = \text{diag}(X)$ (§5.4, p. 39).

- *Failure probability.* $p_n(J)$ is the conditional probability, given J , that the stability implication (5.3) fails at some $t \in [0, T]$ along Y (p. 39).
- *Consistency equations* (4.7), p. 25. For a law P on $\mathbb{R}$: $M = \mathbb{E}_P \tanh y$, $q = \mathbb{E}_P \tanh^2 y$, $d = t + jM$, $s = t + \sigma^2 + jq$; $N(d, s)$ has mean d and variance s . The proof of Lemma 4.2 sets $F(s) = \mathbb{E}_{N(s, s)} \tanh y$ (p. 26), so (4.8) reads $s_* = t + jF(s_*)$, and $q_*(t) = \mathbb{E}_{N(s_*, s_*)} \tanh^2 y$ (4.9).

Role in the argument. Proposition 5.1, the first input in Section 2, asks that $p_n(J) \leq e^{-an}$ for typical J . Its proof bounds the integral (5.5) under the planted law (5.4), excludes unstable approximate zeros at fixed times, passes to the whole path by a mesh, and then undoes the planting (§5.4, p. 37): the planted bound becomes $\mathbb{E}[L(J)p_n(J)\mathbf{1}_{\{\|J\| \leq 2\sqrt{j} + \theta\}}] \leq Ce^{-a_4 n}$ under the original law, and the lower tail of L with Markov’s inequality gives $p_n(J) \leq e^{-an}$ on an event $\mathcal{J}_n$ of probability $1 - o(1)$ (pp. 39–40). Lemma 4.2 identifies the zero set of the scalar rate at $\sigma = 0$ as the curve $P_t = N(s_*(t), s_*(t))$ (after (5.9), p. 29). Strict negativity of the rate away from this curve gives Lemma 5.4(ii), and Lemma 5.5 computes the conditional Hessian at $s = s_*(t)$ (proof, p. 33).

The earlier technique. AMS sample from the SK measure by discretizing the stochastic-localization SDE of a Gibbs sample, with the drift computed by AMP; the analysis runs under a planted model. In their (bold) notation, $\mathbf{A} \sim \text{GOE}(n)$ has entries $N(0, 1/n)$ off the diagonal and $N(0, 2/n)$ on it, $\mu_{\mathbf{A}}(\mathbf{x}) \propto e^{\frac{\beta}{2}\langle \mathbf{x}, \mathbf{A}\mathbf{x} \rangle}$ on $\{-1, 1\}^n$ [5, (1.1), p. 2], and $\mathbf{m}(\mathbf{A}, \mathbf{y})$ is the mean of its tilt by $e^{\langle \mathbf{y}, \mathbf{x} \rangle}$ ((1.2), p. 3).

- *Planted model* (§4.1, p. 10). With $\bar{\nu}$ uniform on $\{-1, 1\}^n$,

$$\mu_{\text{pl}}(d\mathbf{x}, d\mathbf{A}) \propto \exp\left\{-\frac{n}{4}\|\mathbf{A} - \beta\mathbf{x}\mathbf{x}^\top/n\|_F^2\right\} \bar{\nu}(d\mathbf{x}) d\mathbf{A} \quad (4.1).$$

Its $\mathbf{x}$ -marginal is $\bar{\nu}$; given $\mathbf{x}$, $\mathbf{A} = \frac{\beta}{n}\mathbf{x}\mathbf{x}^\top + \mathbf{W}$ with $\mathbf{W} \sim \text{GOE}(n)$ (4.3); $\mu_{\text{pl}}(\cdot \mid \mathbf{A}) = \mu_{\mathbf{A}}$; and $\mu_{\text{pl}}(d\mathbf{A}) = \mu_{\text{GOE}}(d\mathbf{A}) Z_{\text{SK}}(\mathbf{A})$ with $Z_{\text{SK}}(\mathbf{A}) = 2^{-n} \sum_{\mathbf{x}} \exp\{\frac{\beta}{2}\langle \mathbf{x}, \mathbf{A}\mathbf{x} \rangle - \frac{\beta^2 n}{4}\}$ ((4.5)–(4.6)). Footnote 3 (p. 10) compares with the zero-diagonal model.

- *Path laws and likelihood* (pp. 11–12). $\mathbb{Q}$: $\mathbf{A} \sim \mu_{\text{GOE}}$ and $d\mathbf{y} = \mathbf{m}(\mathbf{A}, \mathbf{y}) dt + d\mathbf{B}$ (4.8), a process with the law of $t\mathbf{x} + \bar{\mathbf{B}}(t)$, $\mathbf{x} \sim \mu_{\mathbf{A}}$ ((4.12) and the paragraph after it). $\mathbb{P}$: $\mathbf{x}_0 \sim \bar{\nu}$, $\mathbf{A} \sim \mu_{\text{pl}}(\cdot \mid \mathbf{x}_0)$, $\mathbf{y}(t) = t\mathbf{x}_0 + \mathbf{B}(t)$ (4.9). Proposition 4.2 (p. 11) and (4.14) (p. 12): $\frac{d\mathbb{P}}{d\mathbb{Q}}(\mathbf{A}, \mathbf{y}) = Z_{\text{SK}}(\mathbf{A})$. For $\beta < 1$, the log-normal limit of Z_{SK} (Theorem 4.1, p. 11) and Le Cam’s first lemma make $\mathbb{P}$ and $\mathbb{Q}$ contiguous, and “all results proven under $\mathbb{P}$ transfer to $\mathbb{Q}$ ” (p. 12).
- *State evolution* (pp. 12–15). Under $\mathbb{P}$, the AMP field $\mathbf{z}^k$ of Algorithm 1 ((4.15), p. 12) satisfies: the empirical law of $(x_{0,i} z_i^k)_i$ converges to $N(\tilde{\gamma}_k, \tilde{\gamma}_k)$, $\tilde{\gamma}_k = \gamma_k + t$ (Propositions 4.3–4.4, pp. 12–13), where $\gamma_{k+1} = \beta^2 \mathbb{E} \tanh(\tilde{\gamma}_k + \sqrt{\tilde{\gamma}_k} G)$, $G \sim N(0, 1)$ ((4.16) with Proposition 4.4). Since (4.16) and (4.17) at $j = k$ differ only by $\tanh$ against $\tanh^2$, Proposition 4.4 ($\Sigma_{k+1, k+1} = \gamma_{k+1}$, with $\Sigma_{k, k} + t = \tilde{\gamma}_k$) amounts to $\mathbb{E} \tanh(\tilde{\gamma}_k + \sqrt{\tilde{\gamma}_k} G) = \mathbb{E} \tanh^2(\tilde{\gamma}_k + \sqrt{\tilde{\gamma}_k} G)$, and (4.16) becomes $\gamma_{k+1} = \beta^2(1 - \text{mmse}(\tilde{\gamma}_k))$ (4.20), with $\text{mmse}(\gamma) = 1 - \mathbb{E} \tanh^2(\gamma + \sqrt{\gamma} G)$ (p. 13).
- *Fixed point* (Lemma 4.5, pp. 13–14). (c) $\gamma_* = \beta^2(1 - \text{mmse}(\gamma_* + t))$ (4.21) has a nonnegative solution for $t \geq 0$, unique for $t > 0$ (p. 14), and $\gamma_k \rightarrow \gamma_*$. AMS label the lemma as Lemma 6.1 of Deshpande, Abbe and Montanari [13], and take from there the monotonicity and convexity of mmse (part (a)). The sampler’s constant is $q_* = \lim_k q_k$, $q_{k+1} = \mathbb{E} \tanh^2(\beta^2 q_k + t + \sqrt{\beta^2 q_k + t} G)$ ((2.5), p. 5), so $\gamma_* = \beta^2 q_*$.

Correspondence. Dictionary: $J = \beta\mathbf{A}$ off the diagonal, $j = \beta^2$, $X = \mathbf{x}$ (or $\mathbf{x}_0$), $Y(t) = \mathbf{y}(t)$, and $s = \gamma + t$.

Object	Preprint	AMS [5]
Original law	$J, X \sim \mu_0, Y(t) = tX + B(t)$ (Proposition 5.1, p. 27)	$\mathbb{Q}$ (4.8) in the form (4.12): $t\mathbf{x} + \bar{\mathbf{B}}(t)$, $\mathbf{x} \sim \mu_{\mathbf{A}}$
Likelihood	$L(J) = Z(J)/\mathbb{E}Z(J)$ (p. 39)	$Z_{\text{SK}}(\mathbf{A})$ (4.6)
Size-biased law	density $\propto e^{x^\top Jx/2}$ times the Gaussian density; X uniform; given X, J_{ik} independent with mean jX_iX_k/n , variance j/n (p. 39)	(4.1), (4.3): $\mathbf{x} \sim \bar{\nu}$, $\mathbf{A} = \frac{\beta}{n}\mathbf{x}\mathbf{x}^\top + \mathbf{W}$; $\mu_{\text{pl}}(\cdot \mathbf{A}) = \mu_{\mathbf{A}}$
Diagonal	independent $N(j/n, 2j/n)$ entries added (p. 39)	GOE diagonal plus spike β/n (4.3); footnote 3 (p. 10)
Observation	“Conditional observation paths given J are unchanged by size bias” (p. 39)	$\mathbb{P}$ (4.9); $d\mathbb{P}/d\mathbb{Q}$ depends on $\mathbf{A}$ only (Proposition 4.2)
Gauge	conjugation by D_X gives (5.4) (pp. 28, 39)	$\mathbb{P}$ at time t , conjugated by $\text{diag}(\mathbf{x}_0)$ (first display below)
Transfer	$\mathbb{E}[Lp_n \mathbf{1}] \leq Ce^{-a_4 n}$, (5.27), Markov: $p_n \leq e^{-a_n}$ on $\mathcal{J}_n$ (pp. 39–40)	$\mathbb{E}_{\mathbb{P}} F = \mathbb{E}_{\mathbb{Q}}[Z_{\text{SK}} F]$ (4.14); contiguity (p. 12)
Fixed point	$s_* = t + j\mathbb{E}_{N(s_*, s_*)} \tanh y$ (4.8), p. 25	fixed-point form of (4.16); (4.21) (p. 14)
Nishimori	$\mathbb{E}_{N(s, s)} \tanh y = \mathbb{E}_{N(s, s)} \tanh^2 y$ (4.10), p. 25	Proposition 4.4: (4.16) and (4.17) agree at $j = k$, giving (4.20) (pp. 12–13)
Overlap	$q_*(t) = (s_* - t)/j$ (4.9)	$q_* = \gamma_*/\beta^2$ (2.5)
Equality law	$P_t = N(s_*, s_*)$ (display after (5.9), p. 29)	limit law of $(x_{0,i}z_i^k)_i$ at the fixed point (Propositions 4.3–4.4)

Planted law. With $D = \text{diag}(\mathbf{x}_0)$, AMS’s $\mathbb{P}$ at time t gives

$$\tilde{J} := \beta D A D = \frac{j}{n} \mathbf{1}\mathbf{1}^\top + \beta D W D, \quad h_0 := D \mathbf{y}(t) = t\mathbf{1} + D \mathbf{B}(t),$$

where $\beta D W D$ is GOE of scale j and $D \mathbf{B}(t) \sim \sqrt{t} g$, both independent of $\mathbf{x}_0$: this is the preprint’s (5.4). For the likelihood, with $J = \beta(\mathbf{A} - \text{diag } \mathbf{A})$ and $\mathbb{E}Z(J) = 2^n e^{j(n-1)/4}$,

$$Z_{\text{SK}}(\mathbf{A}) = L(J) e^{\frac{\beta}{2} \text{Tr } \mathbf{A} - \frac{\beta^2}{4}},$$

and the last factor tilts each $\beta A_{ii} \sim N(0, 2j/n)$ to $N(j/n, 2j/n)$, the diagonal the preprint adds. So the preprint’s size-biased law is AMS’s $\mathbb{P}$, its original law is $\mathbb{Q}$, and its planted estimate is (4.14) integrated against $p_n(J) \mathbf{1}_{\{\|J\| \leq 2\sqrt{j} + \theta\}}$.

Fixed point. With $s_* = \gamma_* + t$ and $j = \beta^2$,

$$s_* = t + j \mathbb{E}_{N(s_*, s_*)} \tanh y \iff \gamma_* = \beta^2 \mathbb{E} \tanh(\gamma_* + t + \sqrt{\gamma_* + t} G) = \beta^2(1 - \text{mmse}(\gamma_* + t)),$$

the last equality being the preprint’s Nishimori identity (4.10), which is what AMS Proposition 4.4 asserts through (4.16)–(4.17); in particular $F(s) = 1 - \text{mmse}(s)$ and $s_* = \beta^2 q_* + t$, with q_* from AMS (2.5). The Lipschitz constant in Lemma 4.2, with the same comparison proof, and its upper size bound correspond to AMS (4.24) and to the upper half of (4.23).

What is new in the preprint.

- A quantitative transfer in place of contiguity: $\mathbb{E}Z^2/(\mathbb{E}Z)^2 \leq (1-j)^{-1/2}$, Paley–Zygmund and Gaussian concentration of $\log Z$ give $\mathbb{P}\{L(J) < e^{-\delta n}\} \rightarrow 0$ (5.27), p. 39, and Markov’s inequality converts the exponentially small planted bound into exponentially small conditional failure probabilities $p_n(J)$ for typical J .
- The planted law is used, in the gauge $x = \mathbf{1}$ and with the diagonal kept during the computation and removed afterwards, inside the smoothed Kac–Rice integral (5.5) (finding S1); AMS use it for AMP state evolution.
- The fixed-point equation arises as the equality case of the scalar entropy inequality (Lemma 4.1) under the consistency equations (4.7), and the proof shows that a consistent Gaussian $N(d, s)$ must have $d = s$ (pp. 25–26); in AMS the Nishimori form comes from Bayes optimality (Proposition 4.4).
- A direct proof of the derivative bound $F' \in [0, 1]$, by Gaussian differentiation and the odd-sign inequality (4.11) (p. 25), applied on p. 26; uniqueness then follows by contraction for every $t \geq 0$, including $t = 0$. AMS derive the same bound, as $-\text{mmse}' \in [0, 1]$ (pp. 14–15), from $\text{mmse}'(0) = -1$ and the convexity in Lemma 4.5(a), and prove uniqueness for $t > 0$ by concavity.

How to check.

1. *Planted law.* Put (5.4) (p. 28) and the size-bias paragraph of §5.4 (p. 39) beside AMS (4.1)–(4.6) (p. 10) and (4.9) (p. 11); check the gauge and diagonal identities displayed above.
2. *Likelihood transfer.* Compare the sentence on conditional observation paths, and the planted estimate together with its Markov step (pp. 39–40), against AMS Proposition 4.2 (p. 11), (4.12)–(4.14) and the transfer sentence (p. 12).
3. *Fixed point.* Compare Lemma 4.2, (4.8)–(4.10) (p. 25) with AMS (4.16) (p. 12), Proposition 4.4 and (4.20) (p. 13), (4.21) (p. 14) and (2.5) (p. 5), using $s_* = \gamma_* + t$ and $j = \beta^2$.

4.3 P3. A route to a Poincaré inequality from covariance bounds at stochastic-localization fields, via the inverse TAP Hessian

Summary. El Alaoui and Gaitonde (AG below) [6, §2, pp. 3–4] set out, without proofs, a route, which they call possibly relevant to Poincaré and log-Sobolev inequalities for the SK model at every $\beta < 1$: at fields y produced by stochastic localization, $\text{Cov}(\mu_y)$ should be the inverse of the negative TAP Hessian; its expected operator norm would be bounded given a Frobenius approximation bound and uniform positive definiteness of that matrix outside probability $O(1/n)$; and an operator-norm covariance bound along stochastic localization gives a Poincaré inequality by the technique of Eldan–Shamir and Chen–Eldan. The preprint carries out this route: it proves positivity of exactly this matrix, in symmetrized form, at every approximate TAP root on the observation path (Proposition 5.1, p. 27); at those fields, with the TAP linear-response vector as test vector (§6.4, pp. 45–48), it proves a covariance bound (Proposition 7.2, p. 56) equivalent up to constants to $\|\text{Cov}_{\mu_h}(x)\| \leq C$; and it transfers that bound to μ_0 (§2, pp. 7–10). The preprint cites AG only for their zero-field Theorem 1.1 (§1.1, p. 4); it says neither that AG proposed the route nor that Proposition 7.2 is an operator-norm covariance bound at the posterior fields. Grade: PARTIAL, because what AG contribute is a route, not proofs, and their one concrete step, the Frobenius bound (2.7) with that positivity at the mean of μ_y , is not reused: the preprint never forms $\mathcal{H}\text{Cov} - I$, and its positivity holds at every approximate root outside probability e^{-an} . The transfer step is Eldan–Shamir’s and Chen–Eldan’s (the preprint credits Chen–Eldan), and the linear-response formula Plefka’s and Adhikari et al.’s, as AG say.

Setup. AG write the coupling matrix as βA , with $A_{ii} = 0$ and $A_{ij} = A_{ji} = g_{ij}/\sqrt{n}$ for standard Gaussians g_{ij} [6, Thm 1.3, p. 2], and the external field as y ; so $J = \beta A$, $j = \beta^2$, and AG's μ_y is the preprint's μ_h at $h = y$. They put $Q(m) = \|m\|^2/n$, $D(m) = \text{diag}(1/(1 - m_i^2))$ (pp. 3–4) and, with ent the binary entropy (their h), $\mathcal{F}_{\text{TAP}}(m) = \frac{\beta}{2}\langle m, Am \rangle + \sum_i \text{ent}(m_i) + \frac{n\beta^2}{4}(1 - Q(m))^2$ ((2.3), p. 3). In the preprint, y is instead the coordinate $\text{arctanh } m$ and A a nonnegative diagonal matrix; in the tables below each letter keeps its meaning in its own column. V_y , $\mathbf{H}_J(y, A)$ and $\mathcal{H}(m)$ are as in Section 2. In the implicit construction at a stable field h_0 with root r , $t_* = \tanh r$ (§6.4, pp. 45–47), $m^k = \tanh h^k$ is an iterate of (6.1) near t_* ; A is a positive diagonal within $C\rho\sqrt{n}$ of V_{h^k} in Frobenius norm, with average χ ((6.12), (6.17)); and the test vector w solves the linear system (6.18), whose matrix is $I + AL_2$ with $L_2 = j\chi I - J + \frac{j}{n}(m^{k-1} + t_*)\ell^\top$, $\ell \approx -t_*$.

Role in the argument. This route is the architecture of the proof. Proposition 5.1 gives, before the stopping time of Corollary 5.2, a margin for $\mathbf{H}_J$ at every approximate root; Lemma 6.3 turns it into a bounded inverse for the implicit system (6.18) (proof, p. 47) and the margin (6.15) (p. 46); Proposition 7.2 turns the test vector into the covariance bound (1.5); and §2.2 inserts that bound into $u'(t) = -\mathbb{E}[\mathbf{1}_{\{t < \tau\}} \|\text{Cov}_t(f, x)\|^2]$, giving $u' \geq -C_d(u + d_0)$ and the retained variance (2.4) (p. 8); with the terminal gap and the bad-path bound this gives Theorem 1.1 with $C_\beta = 4e^{C_d T}$ (§2.4, p. 10).

The earlier technique. For zero field and $\beta < 1$, AG [6] (arXiv v3, 2024) prove $\mathbb{E}\|\text{Cov}(\mu)\|_{\text{op}} \leq C(\beta)$ (Thm 1.1, p. 2) by inverting the resolvent bound $\mathbb{E}\|((1 + \beta^2)I - \beta A)\text{Cov}(\mu) - I\|_F^2 \leq C(\beta)$ (Thm 1.3, p. 2) off an event of probability $e^{-c(\varepsilon)n}$ ((1.5)–(1.11), pp. 2–3). Their §2, announced on p. 2, proposes three steps at a nonzero field:

1. *Linear response* ((2.2)–(2.5), p. 3). For a typical field, $\log Z(y) = \max_m \{\mathcal{F}_{\text{TAP}}(m) + \langle y, m \rangle\} + o_{\mathbb{P}}(n)$, and two y -derivatives suggest

$$\text{Cov}(\mu_y) \approx -(\nabla^2 \mathcal{F}_{\text{TAP}}(m))^{-1} = \left(D(m) - \beta A + \beta^2(1 - Q(m))I - \frac{2\beta^2}{n}mm^\top\right)^{-1},$$

with m the maximizer. AG credit the formula, as a heuristic, to Plefka [14] and to [17, (1.12), p. 6] (AG p. 4).

2. *Positivity* (p. 4). For β small enough (e.g. below the AT line when y is Gaussian), one may try to prove the Frobenius bound (2.7) $\mathbb{E}\|(-\nabla^2 \mathcal{F}_{\text{TAP}}(m))\text{Cov}(\mu_y) - I\|_F^2 \leq C(\beta)$, with m the mean of μ_y . A bound on $\mathbb{E}\|\text{Cov}(\mu_y)\|_{\text{op}}$ would follow, by the argument of (1.10), once $\lambda_{\max}(\nabla^2 \mathcal{F}_{\text{TAP}}(m)) \leq -\varepsilon$ outside an event of probability $O(1/n)$. AG note that Celentano's local strong concavity [11] holds only with probability $1 - o_n(1)$, short of that rate.
3. *Transfer* (p. 4). By the technique of Eldan and Shamir [15], generalized by Chen and Eldan [16], operator-norm covariance bounds at the fields visited by Eldan's stochastic localization yield Poincaré and log-Sobolev inequalities for μ . In Chen–Eldan the step is the Cauchy–Schwarz bound $\langle v_t, C_t v_t \rangle \leq \|C_t^{1/2} \text{Cov}(\nu_t) C_t^{1/2}\|_{\text{op}} \text{Var}_{\nu_t} \varphi$ [16, (14), p. 22], integrated along stochastic localization in [16, (20), p. 24].

Correspondence.

Step	AG §2	Preprint
Fields	$\mu_y \propto \exp\{\frac{\beta}{2}\sigma^\top A\sigma + \langle y, \sigma \rangle\}$ (2.1), y from Eldan’s stochastic localization (p. 4)	μ_h at $h = Y(t) = tX + B(t)$ (1.4), p. 6, credited to Eldan and El Alaoui–Montanari (p. 4)
TAP point	Maximizer m of $\mathcal{F}_{\text{TAP}}(m) + \langle y, m \rangle$ ((2.2), p. 3)	Unique zero r of F_{J,h_0} (Lemma 5.7, p. 40); $\tanh r$ is the critical point of the potential Φ of its proof (p. 40) (not the kernel Φ of (6.8)), and $\Phi = -\mathcal{F}_{\text{TAP}} - \langle h_0, \cdot \rangle + n(\log 2 + j/4)$
TAP Hessian	$-\nabla^2 \mathcal{F}_{\text{TAP}}(m)$ ((2.5), p. 3)	$H_J(y, V_y) = V_y^{1/2} \Phi''(m) V_y^{1/2}$ (5.2), p. 27; p. 40
Positivity	$\lambda_{\max}(\nabla^2 \mathcal{F}_{\text{TAP}}(m)) \leq -\varepsilon$ at the mean vector m of μ_y , outside probability $O(1/n)$; Celentano’s $1 - o(1)$ falls short (p. 4)	$H_J(y, A) \succ cI$ at all approximate zeros y of $F_{J,Y(t)}$, diagonals A near V_y and $t \leq T$, outside probability e^{-an} (5.3), p. 27; like AG, the preprint contrasts Celentano’s local convexity with what it needs (AG: a quantitative probability; the preprint: every approximate root along the path), without crediting AG (p. 4)
Linear response	$\text{Cov}(\mu_y) \approx (-\nabla^2 \mathcal{F}_{\text{TAP}}(m))^{-1}$ ((2.4), p. 3), to be made precise by (2.7) (p. 4)	Test vector $w = (I + AL_2)^{-1} Ae$ (6.18), p. 47, which the preprint calls its linear-response test (§6, p. 41)
Covariance bound	$\mathbb{E}\ \text{Cov}(\mu_y)\ _{\text{op}} \leq C$ (p. 4)	$\ \text{Cov}_{\mu_{h_0}}(G, x)\ \leq C(\text{Var } G + \mathcal{D}_{h_0}(G))^{1/2}$ for all G at every stable field (7.2), p. 56; equivalent up to constants to $\ \text{Cov}_{\mu_{h_0}}(x)\ \leq C'$
Transfer	Eldan–Shamir, Chen–Eldan (p. 4)	$u' \geq -C_d(u + d_0)$ from (1.5), then (2.4) (§2.2, p. 8); framework credited to Chen–Eldan (pp. 4, 7)

The two key formulas side by side, with AG’s βA equal to J , $j = \beta^2$ and $m = \tanh y$, so that $b(y) = 1 - Q(m)$ and $V_y^{-1} = D(m)$:

AG	Preprint
TAP Hessian, (2.5), p. 3: $-\nabla^2 \mathcal{F}_{\text{TAP}}(m) = D(m) - \beta A + \beta^2(1 - Q(m))I - \frac{2\beta^2}{n}mm^\top$	Stability matrix (5.2), p. 27 at $A = V_y$: $H_J(y, V_y) = V_y^{1/2}(D(m) + jb(y)I - J - \frac{2j}{n}mm^\top)V_y^{1/2} = V_y^{1/2}(-\nabla^2 \mathcal{F}_{\text{TAP}}(m))V_y^{1/2}$
Linear response, (2.4), p. 3: $\text{Cov}(\mu_y) \approx (-\nabla^2 \mathcal{F}_{\text{TAP}}(m))^{-1}$	Test vector, (6.18), p. 47, source $w_0 = Ae$: $w = A^{1/2}(I + A^{1/2}L_2A^{1/2})^{-1}A^{1/2}e$, with $I + A^{1/2}L_2A^{1/2} = H_J(h^k, A) + O(\rho)$; and $A^{1/2}H_J(h^k, A)^{-1}A^{1/2} = \mathcal{H}(m^k)^{-1}$ at $A = V_{h^k}$

Since $V_y \preceq I$, the margin $H_J(y, V_y) \succ cI$ gives $-\nabla^2 \mathcal{F}_{\text{TAP}}(m) \succeq cD(m) \succeq cI$: AG’s condition with $\varepsilon = c$, at the approximate TAP roots rather than at the mean of μ_y . In the second row, the source $w_0 = Ae$ has $u_0 = e$, $p_0 = 0$ (p. 48), so (6.18) reads $(I + AL_2)w = Ae$, and $(I + AL_2)^{-1}A = A^{1/2}(I + A^{1/2}L_2A^{1/2})^{-1}A^{1/2}$; the matrix the preprint places within $O(\rho)$ of $I + A^{1/2}L_2A^{1/2}$ (p. 47) is $I + jb_kA - A^{1/2}JA^{1/2} - \frac{2j}{n}A^{1/2}m^k(m^k)^\top A^{1/2} = H_J(h^k, A)$, as $b(h^k) = b_k$. Proposition 7.2 is an operator-norm bound: for $G = e^\top x$ with $\|e\| = 1$, $d_i G = e_i$, so $\mathcal{D}_{h_0}(G) = \mathbb{E} \sum_i v_i^1 e_i^2 \leq 1$, and (7.2) gives $\|\text{Cov}(x)e\|^2 \leq C^2(e^\top \text{Cov}(x)e + 1)$, hence $\|\text{Cov}_{\mu_{h_0}}(x)\| \leq C^2 + C$; conversely, Cauchy–Schwarz gives $\|\text{Cov}(G, x)\|^2 \leq \|\text{Cov}(x)\| \text{Var } G$. In §2.2 it enters where Chen–Eldan’s (14) uses $\|\text{Cov}(\nu_t)\|_{\text{op}}$.

What is new in the preprint.

- Positivity at *every* approximate root, for perturbed diagonals, simultaneously for $t \leq T$, outside probability e^{-an} (Proposition 5.1), far stronger than the $O(1/n)$ rate AG name, as the bad-path bound of §2.3 requires; it is proved by a Kac–Rice computation and an approximate-root exclusion (findings S1 and S2).
- A deterministic covariance bound at every stable field, without the resolvent bound (2.7): the preprint tests the TAP residual of a Gibbs-sample-started recursion against w through spin-flip residual identities (Lemmas 6.5–6.6, pp. 53–55; (7.3), p. 57) and the scalar identity (6.9) (finding P1).
- Exceptional paths: the preprint stops at the first failure of stability and bounds the variance on bad paths, for every f , by the f^2 -tilted observation law and Proposition 7.3 (§2.3, pp. 8–9) (finding P4).
- A terminal gap at a large fixed time (Proposition 8.3, p. 63), which AG do not discuss.

How to check.

1. *Matrix.* Put AG (2.3)–(2.5) (p. 3) beside (5.1)–(5.2) (pp. 26–27) and the potential in the proof of Lemma 5.7 (p. 40), with $J = \beta A$, $j = \beta^2$ and AG’s y as h_0 .
2. *Positivity.* Compare the paragraph after AG (2.7) (p. 4) with Proposition 5.1 and Corollary 5.2 (p. 27).
3. *Linear response.* Compare AG (2.4)–(2.5) (p. 3) with the opening of §6 (p. 41) and the inversion of (6.18) in the proof of Lemma 6.3 (p. 47).
4. *Route and credit.* Read AG p. 2 (pointer to §2) and the two paragraphs after (2.7), on Celentano and on Eldan–Shamir (p. 4); then the preprint’s §1.1 (p. 4), Proposition 7.2 (p. 56) and §2.2 (p. 8).

4.4 P4. Stopped stochastic localization with exceptional paths

Summary. Huang, Mohanty, Rajaraman and Wu (HMRW) introduce stopped localization as “a generic tool” for weak Poincaré inequalities: stop a stochastic localization at the first failure of a high-probability path estimate that does not involve the test function, call a path bad if it stops early or if its terminal measure fails the terminal inequality, and split the retained variance into a good part, bounded by the terminal inequality, and a bad part [7, §2, p. 15; §6.2, Lemmas 6.6 and 6.8, pp. 26–27; Lemmas 4.11 and A.6, pp. 20 and 74]. The preprint’s transfer from posterior estimates to Theorem 1.1 has this skeleton: a test-function-independent stopping time $\tau = T \wedge \tau_0$ at the first possible loss of field stability (Corollary 5.2, p. 27; §2.2, p. 7), the bad event with its bound (2.3), stopped variance retention (2.4), and the good/bad split (§2.4, p. 10). The preprint does not cite HMRW. For the framework of its §2 it cites El Alaoui–Montanari for the Gaussian

channel and Chen–Eldan for “the variance/Dirichlet-form framework” (§2, p. 7), and Lehec only for the drift-energy identity (§2.3, p. 8). Chen–Eldan’s annealing theorem gives the good half, with the Dirichlet-form monotonicity (2.2) in the role of their Proposition 48, but assumes the terminal gap almost surely, with no exceptional paths [16, Theorem 46 and Proposition 48, pp. 32–33]; the uncredited content is the stopping-and-exceptional-path scaffold. Grade: PARTIAL, because that scaffold is reused while HMRW’s defining step, charging the bad part by $\text{osc}(f)^2$ times its probability, is replaced by an f^2 -tilted entropy bound.

Setup. Section 2 of this document defines μ_h , $\mathcal{D}_h$ and the observation process $Y(t)$. This finding also needs:

- *Test function.* $\mathbb{E}_{\mu_0} f = 0$, $\mathbb{E}_{\mu_0} f^2 = 1$ and $d_0 = \mathcal{D}_0(f)$. With $\mathbb{E}_t$ the expectation under $\mu_{Y(t)}$, $M_t = \mathbb{E}_t f$, $N_t = \mathbb{E}_t f^2$ and $V_t = N_t - M_t^2 = \text{Var}_{\mu_{Y(t)}}(f)$ (§2.2, p. 8).
- *Stopping.* τ_0 is the first $t \leq T$ with $d_J(Y(t)) \leq \rho_0$, where $d_J(h)$ is the infimum of $\|F_{J,h}(y)\|/\sqrt{n}$ over pairs (y, A) with A near V_y and $H_J(y, A) \neq cI$. Then $\mathbb{P}(\tau_0 \leq T \mid J) \leq e^{-an}$ on the matrix event of Proposition 5.1 (Corollary 5.2 and its proof, p. 27). Set $\tau = T \wedge \tau_0$.
- *Bad paths.* With the terminal set $\mathcal{T}_n(J) = \{h : \text{Var}_{\mu_h}(g) \leq 2\mathcal{D}_h(g) \text{ for every } g\}$, which $Y(T)$ misses with probability at most e^{-cn} (Proposition 8.3, (8.14), p. 63), a path is bad if $\tau_0 \leq T$, or if $\tau_0 > T$ and $Y(T) \notin \mathcal{T}_n(J)$. Then $\mathbb{P}(\text{bad}) \leq 2e^{-an}$ (2.3); on good paths $\tau = T$.
- *Tilted observation law.* $\mathbb{P}^f$ is the law of the observation path when $X \sim f^2\mu_0$; its density on $\mathcal{F}_\tau$ is N_τ (§2.3, p. 8).

Role in the argument. The preprint’s §2 uses this skeleton to carry back to μ_0 , uniformly over f , the three inputs of its §1.4 (listed in Section 2). Before τ the field is stable, so the directional covariance bound holds for every f and gives $u' \geq -C_d(u + d_0)$ for $u(t) = \mathbb{E}V_{t \wedge \tau}$, hence (2.4). The terminal gap on good paths and $\mathbb{P}^f$ on bad ones then give $d_0 \geq v/4$ and $C_\beta = 4e^{C_d T}$ (§2.4, p. 10). The introduction (§1.1, p. 4) motivates the f^2 -weighting by the variance that an exponentially rare observation event can carry: this scaffold’s bad part.

The earlier technique. HMRW [7] (arXiv v2, November 2024) use stochastic localization with driving matrix C_t , $\mu_t(x) \propto \mu_0(x) \exp(-\frac{1}{2}\langle x, \Sigma_t x \rangle + \langle y_t, x \rangle)$, $\Sigma_t = \int_0^t C_s^2 ds$ (Definition 6.2, p. 25). A $(\rho_{\text{PI}}, \varepsilon)$ -weak Poincaré inequality for Glauber dynamics is $\text{Var}_\pi[f] \leq \rho_{\text{PI}}^{-1} \mathcal{E}(f, f) + \varepsilon \text{osc}(f)^2$ (Definition A.1, p. 73).

- *Stopping* (§6.2, p. 26). Approximate conservation of variance can be ensured by stopping “the localization scheme whenever it fails to hold”. Lemma 6.6: for every φ ,

$$\tau = T \wedge \inf\{t : \|C_t^{1/2} \text{Cov}(\mu_t) C_t^{1/2}\|_{\text{op}} \geq K\} \implies \mathbb{E} \text{Var}_{\mu_\tau}[\varphi] \geq e^{-TK} \text{Var}_\mu[\varphi].$$

Remark 6.7 notes that for variance retention alone the localization may depend on φ , and extends the stopping to any rate-of-decay condition, “by stopping the localization process whenever these conditions fail to hold”. Lemma 6.8 uses the φ -independent τ of Lemma 6.6.

- *Exceptional paths* (Lemma 6.8, p. 27). If with probability $1 - \eta_1$ the covariance bound holds for all $t \in [0, T]$, and with probability $1 - \eta_2$ the measure μ_T satisfies a $(\rho_{\text{PI}}, \delta)$ -weak Poincaré inequality, then μ satisfies a $(\rho_{\text{PI}} e^{-TK}, e^{TK}(\delta + \eta_1 + \eta_2))$ -weak Poincaré inequality. Proof: use the decomposition $\mu = \mathbb{E}\mu_\tau$; $\tau = T$ with probability $1 - \eta_1$, so μ_τ satisfies the terminal inequality with probability at least $1 - \eta_1 - \eta_2$; conclude by Lemma 4.11.

- *Good/bad split* (Lemma 4.11, p. 20; Glauber version Lemma A.6, p. 74, which uses Fact A.3, p. 73: $\mathcal{E}_\pi(f, f) \geq \mathbb{E}_z \mathcal{E}_{\pi_z}(f, f)$). If $\mathbb{E}_z \text{Var}_{\pi_z}[f] \geq C_{\text{Var}} \text{Var}_\pi[f]$ for all f and π_z satisfies a $(\rho_{\text{PI}}, \delta)$ -weak Poincaré inequality with probability $1 - \eta$, then π satisfies a $(\rho_{\text{PI}} C_{\text{Var}}, (\delta + \eta)/C_{\text{Var}})$ -weak one. The proof calls z good if π_z satisfies the inequality, applies it inside $\mathbb{E}[\mathcal{E}_{\pi_z}(f, f) \mathbf{1}_{\text{good}}]$, and bounds $\mathbb{E}[\text{Var}_{\pi_z}[f] \mathbf{1}_{\text{not good}}]$ by $\eta(\text{osc}(f)^2 + \sup \|\nabla f\|^2)$, which is $\eta \text{osc}(f)^2$ in the Glauber form (Lemma A.6).
- *Application* (§7, p. 29). For spherical spin glasses (driving matrix Id, large constant T), a high-probability covariance bound along the path (Lemma 7.6) and an endpoint weak Poincaré inequality (Lemma 7.7) are plugged into Lemma 6.8.

Davies, Lee, Sandhu and Shi apply these lemmas, with credit to HMRW, to Glauber dynamics for SK [9, Lemma 2.10, p. 9; Theorem 5.5, p. 22] (see finding P6).

Correspondence.

Step	Preprint	HMRW [7]
Localization	$Y(t) = tX + B(t)$, posterior $\mu_{Y(t)}$ (1.4)	driving matrix Id (Def. 6.2; §7, p. 29)
Stopping	$\tau = T \wedge \tau_0$, τ_0 the first t with $d_J(Y(t)) \leq \rho_0$; “does not use the test function” (Cor. 5.2; §2.2)	$\tau = T \wedge$ first t with $\ C_t^{1/2} \text{Cov}(\mu_t) C_t^{1/2}\ _{\text{op}} \geq K$ (Lemma 6.6; proof of Lemma 6.8)
Path estimate	for $J \in \mathcal{J}_n$, $\mathbb{P}(\tau_0 \leq T \mid J) \leq e^{-an}$ (Prop. 5.1, Cor. 5.2)	bound on $[0, T]$ with probability $1 - \eta_1$ (Lemma 6.8); for typical H_N , with probability $1 - e^{-cN^{1/5}}$ (Lemma 7.6)
Retention	$\mathbb{E}V_\tau \geq v - (1 - v)d_0$, $v = e^{-C_d T}$ (2.4)	$\mathbb{E} \text{Var}_{\mu_\tau}[\varphi] \geq e^{-TK} \text{Var}_\mu[\varphi]$ (Lemma 6.6)
Terminal inequality	$\text{Var}_{\mu_{Y(T)}}(f) \leq 2\mathcal{D}_{Y(T)}(f)$ for every f , except with probability e^{-cn} (Prop. 8.3)	μ_T satisfies a weak Poincaré inequality with probability $1 - \eta_2$ (Lemmas 6.8, 7.7)
Bad paths	$\{\tau_0 \leq T\} \cup \{\tau_0 > T, Y(T) \notin \mathcal{T}_n(J)\}$, $\mathbb{P}(\text{bad}) \leq 2e^{-an}$ (2.3)	$\tau < T$ or μ_T fails; probability $\leq \eta_1 + \eta_2$ (proof of Lemma 6.8)
Good part	$\mathbb{E}[\mathbf{1}_{\text{good}} V_\tau] \leq 2\mathbb{E}\mathcal{D}_{Y(T)}(f) \leq 2d_0$ (§2.4), the last step by $\mathbb{E}\mathcal{D}_{Y(t)}(f) \leq \mathcal{D}_0(f)$ (2.2)	$\mathcal{E}_\pi(f, f) \geq \rho_{\text{PI}} \mathbb{E}[\text{Var}_{\pi_z}[f] \mathbf{1}_{\text{good}}] - \delta \rho_{\text{PI}} \text{osc}(f)^2$ (proof of Lemma 4.11, in its Glauber form, Lemma A.6; first step by Fact A.3)
Bad part (not reused)	$\mathbb{E}[\mathbf{1}_{\text{bad}} V_\tau] \leq \mathbb{E}[\mathbf{1}_{\text{bad}} N_\tau] = \mathbb{P}^f(\text{bad}) \leq v/4$ (§2.4, with (2.9))	$\mathbb{E}[\text{Var}_{\pi_z}[f] \mathbf{1}_{\text{not good}}] \leq \eta \text{osc}(f)^2$ (proof of Lemma 4.11, in its Glauber form, Lemma A.6)
Conclusion	$\text{Var}_{\mu_0} f \leq C_\beta \mathcal{D}_0(f)$ (Theorem 1.1), with $C_\beta = 4e^{C_d T}$ (§2.4, p. 10)	$(\rho_{\text{PI}} e^{-TK}, e^{TK}(\delta + \eta_1 + \eta_2))$ -weak Poincaré (Lemma 6.8)

With $\pi_z = \mu_\tau$, $\eta = \eta_1 + \eta_2$ and $C_{\text{Var}} = e^{-TK}$, HMRW’s chain in the Glauber form (first line) and the preprint’s (second line) read

$$\begin{aligned} \mathcal{E}_\mu(f, f) &\geq \rho_{\text{PI}} \mathbb{E}[\text{Var}_{\mu_\tau}[f] \mathbf{1}_{\text{good}}] - \delta \rho_{\text{PI}} \text{osc}(f)^2, & \mathbb{E}[\text{Var}_{\mu_\tau}[f] \mathbf{1}_{\text{bad}}] &\leq (\eta_1 + \eta_2) \text{osc}(f)^2; \\ 2d_0 &\geq 2\mathbb{E}\mathcal{D}_{Y(T)}(f) \geq \mathbb{E}[\mathbf{1}_{\text{good}} V_\tau], & \mathbb{E}[\mathbf{1}_{\text{bad}} V_\tau] &\leq \mathbb{E}[\mathbf{1}_{\text{bad}} N_\tau] = \mathbb{P}^f(\text{bad}), \end{aligned}$$

each closed by its retention bound (Lemma 6.6, resp. (2.4)), where $V_\tau = \text{Var}_{\mu_{Y(\tau)}}(f)$ plays the role of $\text{Var}_{\mu_\tau}[f]$. The good halves coincide (a true Poincaré inequality is the case $\delta = 0$); the bad halves differ.

What is new in the preprint.

- The bad part is bounded under the f^2 -tilted observation law, not by $\text{osc}(f)^2$: $\mathbb{P}^f(\text{bad}) \leq v/4$ uniformly over normalized f with $d_0 \leq 1$, from the stopped drift-energy identity (2.5), the square-reweighted mean estimate (1.6) (Proposition 7.3) applied before τ , the weighted Dirichlet identity (2.6), the entropy bound (2.7) and the log-sum inequality (§2.3, pp. 8–9). This is what yields a Poincaré inequality rather than a weak one.
- Variance retention through $u' \geq -C_d(u+d_0)$, driven by $\|\text{Cov}_{\mu_h}(G, x)\|^2 \leq C(\text{Var}_{\mu_h}(G) + \mathcal{D}_h(G))$ and (2.2), in place of a covariance cap with retention e^{-TK} .
- The stopping criterion is field stability, the least normalized residual $d_J(Y(t))$ of an unstable witness (y, A) , whose rarity is proved by a Kac–Rice-type Gaussian integral (Proposition 5.1; finding S1), not a covariance norm.
- Heat-bath Glauber dynamics on $\{-1, 1\}^n$ for SK at every $\beta < 1$, with a true terminal Poincaré inequality at a large fixed time (Proposition 8.3).

How to check.

1. *Stopping rule.* Corollary 5.2 and its proof (p. 27) and the first paragraph of §2.2 (p. 7) against HMRW Lemma 6.6, the sentence before Lemma 6.5 and Remark 6.7 (p. 26).
2. *Bad event.* §2.2, (2.3), p. 7 against the proof of HMRW Lemma 6.8 (p. 27) and their §2 (p. 15); compare Proposition 5.1 (p. 27) and Proposition 8.3 (p. 63) with HMRW Lemmas 7.6–7.7 (p. 29).
3. *Good/bad split.* The two displays in §2.4 (p. 10) against the proof of HMRW Lemma A.6 (p. 74), that is, of Lemma 4.11 (p. 20) with Fact A.3: the good halves match, (2.2) playing Fact A.3; the bad halves differ.
4. *Cited framework.* Chen–Eldan Theorem 46 (p. 32) and its proof from Proposition 48 (p. 33), the counterpart of (2.2): a stopping time is allowed, and the terminal spectral gap is assumed almost surely.

4.5 P5. Resolvent estimates uniform over adaptive diagonals by metric projection, concentration and chaining

Summary. The preprint’s Lemma 3.1 bounds $\|\text{diag}(Q - P(Q))\|_2$, with $P(Q)$ a noncrossing prediction, by a constant ($O(n^{-1/2})$ normalized) simultaneously over words Q in a GOE matrix W (or J) with at most one inverse $K_A = (I - AW + B_A A)^{-1}$, $B_A = jn^{-1} \text{Tr } A$, and over diagonal parameters that “may be chosen after the random matrix has been observed” (Lemma 3.1 and (3.7), p. 12). Its proof (§3.3, pp. 17–19) follows the assembly that Davies, Lee, Sandhu and Shi use in Theorem B.9 for the same uniformity over all diagonals: metric projection of the GOE onto an operator-norm ball, increments regular in the diagonal parameters (Hölder- $\frac{1}{2}$ in the preprint), Gaussian concentration, and chaining at the natural scale [8, p. 18; Lemma B.8, p. 159; Theorem B.9, pp. 160–162; Corollary B.35, p. 179]. The preprint does not cite Davies et al.; in §3 it cites only Speicher and Vitale, for other points (pp. 11–12). Grade: PARTIAL, because the assembly of standard steps in Theorem B.9 of Davies et al. is followed for the same purpose, while the objects controlled differ and the mean computation is the preprint’s own.

Correspondence. With the normalized GOE A of Davies et al., the preprint’s W is βA ; for its diagonal $A = D^{-1}$, $K_A = \bar{Q}D$ with $\bar{Q} = (B_A I - \beta A + D)^{-1}$. Davies et al. bound the diagonal of $D\bar{Q}^2 D = K_A^\top K_A$ (two inverse factors), Lemma 3.1 words with one. Neither contains the other.

Step	Davies et al. [8]	Preprint [10]
Statement	Cor. B.35 ($\beta < 1/2$, $\gamma = (1 - 2\beta)/2$): with probability $1 - e^{-O((\log n)^\xi)}$, for all $D \succeq I$, $a \in [0, 1]$, $\ E_{\mathcal{D}_n}[D(aI - \beta A + D)^{-2}D] - F'(a)D(F(a)I + D)^{-2}D\ _{L^2} \leq C\beta/(\gamma^4\sqrt{n})$	(3.7), $j < 1$: with probability $1 - Ce^{-c_1 n}$, for all words Q of length $\leq L$ with at most one $K_A(J)$ and all $0 \leq A \leq a_+ I$ with $S_A(1; J) \succeq cI$, $\ \text{diag}(Q - P(Q))\ _2 \leq C$
Truncation	$\bar{A} = \Pi_{C_{\text{op}}}(A)$, metric projection in $\ \cdot\ _{L^2}$ onto $\{\ M\ _{\text{op}} \leq C_{\text{op}}\}$, 1-Lipschitz (Thm B.9)	$\widehat{W} = \Pi_R(W)$, metric projection in Frobenius norm onto $\{\ M\ \leq R\}$, nonexpansive; also a cutoff $f(S_A)$ (pp. 13–14)
Increments	$\ \nabla\Phi_\theta - \nabla\Phi_{\tilde{\theta}}\ _{L^2} \leq \beta\rho(\theta, \tilde{\theta})$ on the ball, for $\Phi_\theta = \text{tr}_n(TD(aI - \beta A + D)^{-2}D)$, $\theta = (a, D^{-1}, T)$ (Lemma B.8); global via $\Pi_{C_{\text{op}}}$ (p. 161)	$\text{Lip}_F(\widehat{Q}_\theta - \widehat{Q}_{\theta'}) \leq C\sqrt{\Delta}$ if $\ \theta - \theta'\ _\infty \leq \Delta \leq 1$, on the ball, then via Π_R (3.12), p. 14
Chaining	Herbst for increments; γ_2 and Dudley’s integral over $2n + 1$ parameters: $\mathbb{E} \sup_\theta Z_\theta \leq C\beta/(\gamma^4\sqrt{n})$, and w.h.p.; duality in T (pp. 161–162)	Gaussian scale $C\sqrt{\Delta/n}$; meshes of spacing $2^{-\ell}$, union bound at $C'\sqrt{\ell + 1}2^{-\ell/2}$ per level: $\sup_\theta N_k(\widehat{Q}_\theta - \mathbb{E}\widehat{Q}_\theta) \leq C$, probability $1 - Ce^{-c_1 n}$ (3.22), p. 18

How to check.

1. *Increments.* Pages 13–14, to (3.12), against Lemma B.8 and Theorem B.9 (pp. 159–161).
2. *Chaining.* Proof of Lemma 3.1 to (3.22) (pp. 17–18) against that of Theorem B.9 (pp. 161–162).
3. *Objects and means.* Lemma 3.1 (p. 12) against Corollary B.35 (p. 179); means by loop equations (Lemma 3.3, pp. 15–17) and by free interpolation (Proposition B.23, p. 169).

4.6 P6. A weak Poincaré inequality for SK heat-bath Glauber dynamics at $\beta < 1/2$ via approximate stochastic localization

Summary. Davies, Lee, Sandhu and Shi (DLSS; arXiv v1, July 9, 2026) proved by approximate stochastic localization that for every $\beta < 1/2$ and $\delta > 0$, with probability at least $1 - \delta$ for large n , heat-bath Glauber dynamics for the zero-field SK model satisfies the weak Poincaré inequality $\text{Var}_{\mu_0}(f) \leq e^{C/\epsilon}\mathcal{D}_0(f) + \epsilon \text{osc}(f)^2$ for all $\epsilon \in (0, 1)$ [9, Cor. 1.5, p. 2], the first functional inequality for this chain past the threshold $\beta \approx 0.295$ of Anari, Koehler and Vuong (DLSS, p. 1). It is a weaker version, on half the range, of the preprint’s Theorem 1.1 (p. 3), which implies it for $C \geq \max(\log C_\beta, 0)$. DLSS was posted before both works the preprint cites for $\beta < 1/2$, Wang (cited version August 28, 2026) and Boban, Li and Oveis Gharan (BLO; September 11, 2026), whose results also imply it on its range: BLO’s constant unscaled gap [18, Thm. 1.5, Cor. 1.6, p. 5] and the Poincaré inequality behind Wang’s cited Theorem 1.1 [19, Thm. 3.1, p. 8]. The preprint does not cite DLSS: its history passes from Anari, Koehler and Vuong directly to Wang and BLO (§1.1, p. 4), both of which discuss DLSS [19, p. 5], [18, p. 7]. Grade: PARTIAL (B): a weaker, partial-range version of the main theorem, with none of its proof reused.

Correspondence.

	Preprint	DLSS [9]
Chain	heat-bath, form $\mathcal{D}_0$ (1.1), p. 3	Glauber, $\mathcal{E}(f, f) = \langle f, (I - P)f \rangle_\mu = n^{-1} \mathcal{D}_0(f)$ (Def. 2.1, p. 6)
On μ_0	$\text{Var}_{\mu_0}(f) \leq C_\beta \mathcal{D}_0(f)$, $\beta < 1$, probability $\rightarrow 1$ (Thm. 1.1, p. 3)	a $(\frac{1}{n}e^{-C/\epsilon}, \epsilon)$ -WPI, $\epsilon \in (0, 1)$, $\beta < \frac{1}{2}$, probability $\geq 1 - \delta$ (Cor. 1.5, p. 2)
Posterior at time T	$\text{Var}_{\mu_Y}(f) \leq 2\mathcal{D}_Y(f)$, any law of X , $\beta < 1$, outside probability e^{-cn} , one fixed c (Prop. 8.3, p. 63)	$\mu_{\beta A, y_T}$: $\text{Var} \leq c(\beta)^{-1} \mathcal{D} + e^{-Cn} \text{osc}^2$, X Gibbs, $\beta < \frac{1}{2}$, outside probability e^{-Cn} , any fixed C (Thm. 5.1, p. 21): weaker inequality, but a rate as large as wanted
Back to μ_0	stopping without f ; gap on good paths; bad paths under the f^2 -tilted law (§§2.2–2.4, pp. 7–10)	WPI decomposition (Cor. 3.2, p. 16); proximal-sampler WPI from s -conductance, $s = \epsilon$ (Thm. 4.4, p. 18; Lemma 4.5, p. 20)

How to check.

1. DLSS Corollary 1.5 (p. 2), Definitions 2.1 and 2.3 (p. 6) and $\text{Err} = \text{osc}^2$ (p. 7) against (1.1) and Theorem 1.1 (p. 3): for $\epsilon < 1$, $e^{C/\epsilon} \geq e^C \geq C_\beta$.
2. The history (§1.1, p. 4) against DLSS's date and p. 1, Wang (Thm. 3.1, p. 8; p. 5) and BLO (Cor. 1.6, p. 5; p. 7).
3. DLSS Corollary 3.2 (p. 16), Theorem 4.4 (p. 18) and Theorem 5.1 (p. 21) against §§2.2–2.4 (pp. 7–10) and Proposition 8.3 (p. 63): no step is shared. DLSS use the stopping scaffold of finding P4 only inside Theorem 5.1 (their Theorem 5.5, p. 22).

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